

TRDP II

Explaining BERT, CLIP Text and CLIP Vision with Partition SHAP

A qualitative and quantitative evaluation of text and image explanations

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This report continues our previous report, in which we explained the sentiment decisions of a BERT classifier with Partition SHAP. We extend that work in three directions: we explain both sentiment outputs instead of a single score, we apply the same method to the text and vision encoders of CLIP, and we evaluate all explanations with established faithfulness and plausibility metrics. The BERT checkpoint, the qualitative sentence set and the label convention of the previous report were kept, and all evaluations were computed on top of the stored explanations without changing them.

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Abstract

In our previous report we applied Partition SHAP to a BERT classifier fine-tuned on SST-2 and showed that it recovers the expected sentiment words for ten qualitative sentences. In this report we extend that analysis to three models: the same BERT classifier, the frozen CLIP text encoder used as a zero-shot sentiment classifier through prompt prototypes, and the CLIP vision encoder.

We first explain the negative and positive output of BERT and CLIP text for the ten qualitative sentences (S1–S10). Both models highlight the obvious sentiment words, but BERT does so with clear margins while CLIP’s decisions are very close and partly driven by neutral words such as *movie* or *film*. We then evaluate the explanations on 500 balanced SST-2 sentences. The explanations of BERT are more faithful than those of CLIP text (comprehensiveness 0.772 vs. 0.705, sufficiency 0.223 vs. 0.331, deletion AOPC 0.890 vs. -0.368) and agree better with human word-level sentiment ratings (rationale F1 0.494 vs. 0.350, using the Stanford Sentiment Treebank word ratings as a proxy reference). For CLIP vision on ten images, removing the highest-ranked patches quickly lowers the similarity to the original image embedding (deletion AUC 0.711, insertion AUC 0.849, AOPC 0.199), and the highest-ranked patch lies on the main object in 9 of 10 images (7 of 10 without a pixel tolerance). All explanations reproduce the explained model outputs up to floating-point precision. We discuss how the masking baseline, the very small CLIP decision margins and the proxy references limit how far these numbers can be interpreted.

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1 Introduction

The goal of our TRDP work is to understand *which parts of the input* a model uses when it makes a decision. For a sentiment classifier these are the words that make a review positive or negative; for an image model they are the regions of the picture. An explanation of this kind is only useful if it really reflects the model, so an important part of the work is to check the explanations and not only to look at them.

In our previous report we explained a BERT classifier fine-tuned on SST-2 with Partition SHAP [1]. SHAP does not look inside the network. It asks a counterfactual question instead: how does the model output change when some words are hidden and others remain visible? We explained the difference between the positive and the negative logit for ten qualitative sentences (S1–S10). All ten predictions were correct, the base value plus the SHAP values reconstructed the margin exactly, and the strongest attributions went to clear sentiment words such as *beautiful*, *outstanding*, *boring*, *weak* and *captivating*. Masking the highest-ranked words lowered the decision score by 5.31 logits on average, while keeping only those words lost 0.89 logits.

That report left three questions open. First, it explained only one number, the margin between the two classes, so it did not show how each word acts on the NEG and on the POS output. Second, it covered only one model. Third, its faithfulness test used only ten sentences and raw logit values, which cannot be compared across models. The objectives of the present report are therefore:

- (1) to explain the NEG and the POS output of BERT separately and to relate the result to the margin explanation of the previous report;
- (2) to apply the same procedure to the CLIP text encoder [7], used as a zero-shot sentiment classifier through prompts, and to compare its explanations with those of BERT;
- (3) to measure the faithfulness of the BERT and CLIP text explanations on 500 SST-2 sentences with comprehensiveness, sufficiency and deletion AOPC;
- (4) to measure how well the words selected by SHAP agree with human word-level sentiment (token-level rationale precision, recall and F1);
- (5) to explain the CLIP vision encoder at the level of image patches and to evaluate these explanations with deletion and insertion AUC, AOPC and the pointing game.

An important principle throughout the project was that the evaluation must not change the thing it evaluates. The models, datasets, masking procedures and stored SHAP values of the underlying experiments were reused as they were; the metrics were computed on top of them, and every evaluation run checked that the source folders were left untouched.

The report is organized as follows. Section 2 describes the models, the explanation method and the evaluation measures. Section 3 summarizes the implementation workflow and Section 4 the data and experimental setup. Section 5 presents the results, which we discuss in Section 6 before concluding in Section 7. All sentence-level and image-level plots are collected in the appendices.

2 Model and explanation methodology

2.1 SHAP and Partition SHAP

SHAP explains one prediction of a model by distributing the model output over the input features [1]. A feature can be a word, a sub-word token or an image patch. To measure the contribution of a feature, SHAP evaluates the model on many *perturbed* inputs in which some features are hidden.

Hidden text tokens are replaced by a mask token or deleted, and hidden image regions are replaced by a blurred version of the image. The result is one signed value per feature: a positive value means the feature increases the explained output, a negative value means it decreases it.

The most useful property of SHAP for our purposes is *additivity*. For every input x with features $1, \dots, M$ the explanation satisfies

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i, \quad (1)$$

where $f(x)$ is the explained model output, ϕ_0 is the *base value* (the output when all features are hidden) and ϕ_i is the contribution of feature i . We check this equation for every explanation in the report; the difference between the two sides is called the *additivity residual*.

Computing exact Shapley values requires all 2^M subsets of features and is too expensive. As in the previous report, we use the *Partition* explainer of the SHAP library [2], which groups the features into a hierarchy (neighboring words, or neighboring image blocks) and only evaluates coalitions that respect this hierarchy. The resulting values are Owen values, an approximation of Shapley values that still satisfies Equation (1). In all experiments we allowed at most 500 model evaluations per sentence or image.

2.2 BERT sentiment classifier

We use the same checkpoint as in the previous report, `textattack/bert-base-uncased-SST-2` [3, 4], a BERT-base model that was already fine-tuned on the binary SST-2 sentiment task [5, 6]. It outputs two logits, one for the negative (NEG) and one for the positive (POS) class, and predicts the class with the larger logit. We did not train or fine-tune it further. The label convention of the previous report is kept: index 0 is NEG and index 1 is POS. This mapping was re-checked on the ten qualitative sentences before any evaluation.

2.3 CLIP text encoder as a zero-shot sentiment classifier

CLIP [7] is trained to match images with their captions; its text encoder maps a sentence to an embedding vector. It has no sentiment output, so we turn it into a zero-shot sentiment classifier. We write a small set of negative prompts (e.g. “This movie is terrible.”) and positive prompts (e.g. “This movie is excellent.”), embed them, and average the embeddings per class into a NEG and a POS *prototype*. The NEG and POS scores of a sentence are its cosine similarities to the two prototypes, and the predicted class is the one with the higher similarity. These scores are similarities, not probabilities, and they are typically very close to each other. We use the checkpoint `openai/clip-vit-base-patch32` and keep it frozen.

2.4 CLIP vision encoder

The vision tower of the same CLIP checkpoint is a Vision Transformer [8] that splits the 224×224 input into $7 \times 7 = 49$ patches of 32×32 pixels. Like the text encoder, it outputs an embedding and no class. In our vision experiment the explained output is therefore image-only: the cosine similarity between the embedding of a perturbed image and the embedding of the original image. The original image scores exactly 1, and a patch with a large SHAP value is a region that CLIP needs to keep its representation of the image. No text, prompts or labels are involved.

2.5 Evaluation of explanations

An explanation can be evaluated in two different ways [9]. *Faithfulness* asks whether the explanation reflects what the model actually does: if we remove the features that the explanation calls important, the model output should change a lot; if we keep only those features, it should change little. *Plausibility* asks whether the explanation agrees with what a human would point at, for example the sentiment words of a review or the object in a photo.

For text we use the faithfulness metrics *comprehensiveness* and *sufficiency* from the ERASER benchmark [9], which the previous report already used on ten sentences, and the *area over the perturbation curve* (AOPC) [10]. For plausibility we use *token-level rationale F1*, which compares the selected words with human-marked words [9]. For images we use the *deletion* and *insertion* curves of RISE [11], AOPC, and the *pointing game* [12], which checks whether the most important location lies on the annotated object. The next two subsections describe the exact procedures.

2.6 Text measures

All four text measures use the same procedure for BERT and CLIP text, so any difference comes from the models and not from the evaluation. They are computed for the decision margin POS–NEG, because it is the quantity that decides the prediction (POS if the margin is positive, NEG otherwise).

Decision score. Removing a word should be judged from the point of view of the predicted class. As in the previous report we therefore use the *decision score*

$$s(x) = \begin{cases} \text{margin}(x) & \text{if the model predicts POS,} \\ -\text{margin}(x) & \text{if the model predicts NEG,} \end{cases} \quad (2)$$

which is positive for the original sentence and shrinks when evidence for the prediction is removed.

Ranking and cut-offs. The content words of a sentence (punctuation excluded) are ranked by the absolute SHAP value of the margin; ties are broken by position. We then take the top 10 %, 20 %, 30 % and 50 % of the content words (at least one word), the same fractions as in the previous report. On average this is 2.1, 3.8, 5.5 and 8.6 words.

The measures.

- **Comprehensiveness** (↑). Delete the top-ranked words and measure how much the decision score drops. A large drop means the explanation found words the model really needs. We average the drop over the four cut-offs.
- **Sufficiency** (↓). Keep *only* the top-ranked words (and the punctuation) and measure how much the decision score drops. A small drop means the selected words alone are enough to support the decision. Again averaged over the cut-offs.
- **Deletion AOPC** (↑). The same deletions as for comprehensiveness, but each drop is divided by the sentence’s own decision score before averaging. This is the measure of the original 500-sentence experiment, which we reproduce exactly.
- **Token-level rationale precision, recall and F1** (↑). Compare the top-ranked words with a set of reference sentiment words: precision is the share of selected words that are reference words, recall is the share of reference words that were selected, and F1 combines both. Averaged over the four cut-offs.

Making BERT and CLIP comparable. In the previous report comprehensiveness and sufficiency were reported in logits, which is fine for one model but not for a comparison. BERT’s margin is measured in logits (on average 6.06), CLIP’s in cosine similarity (on average 0.0046). A raw drop of 1.0 would be huge for CLIP and small for BERT. We therefore divide comprehensiveness and sufficiency by the model’s *average* absolute decision score over the 500 sentences, so that both measures read as “share of a typical decision margin”. Deletion AOPC instead divides by the margin of each individual sentence, as in the original 500-sentence experiment. This makes a difference for CLIP: 73 of its 500 sentences have a margin below 0.001, and dividing by such a tiny number produces very large values. We come back to this in Section 5.3.

Reference words for rationale F1. Rationale F1 normally requires humans to mark the words that justify the label of each sentence. SST-2 does not contain such rationales, and we did not want to invent them. The Stanford Sentiment Treebank [5], from which SST-2 is derived, does however contain a human sentiment rating (0 = very negative, 1 = very positive) for every phrase and every single word. We use these ratings as a *proxy* reference: a content word counts as a sentiment word if its rating lies outside SST’s own neutral interval (0.4, 0.6]. All 8,404 content words of the 500 sentences received a rating, and 1,365 of them are sentiment words. Thirty-two sentences contain no sentiment word at all; recall is undefined for them, so they are excluded from the rationale measures for both models (468 sentences remain). Because the words were rated out of context, this measures agreement with human lexical sentiment rather than with sentence-specific human rationales.

2.7 Image measures

The image measures reuse the ranking and masking of the vision experiment. The 49 patches are ranked by their absolute SHAP value (all patch values were non-negative, so this is the same as the signed ranking). A removed patch is replaced by the blurred image that SHAP used as its background, so the explanation and its evaluation share one definition of a “missing” region. The fully blurred image scores between 0.30 and 0.56 (mean 0.45), not zero.

- **Deletion AUC** ($\downarrow$). Blur the patches one at a time, from the most to the least important, and record the similarity after each step. The deletion curve starts at 1 (original image) and ends at the fully blurred score. We report the area under this curve with the fraction of removed patches on the x-axis. A good explanation makes the curve fall early, so a lower area is better.
- **Insertion AUC** ($\uparrow$). Start from the fully blurred image and restore the patches in the same order. A good explanation brings the similarity back to 1 quickly, so a higher area is better.
- **AOPC** ($\uparrow$). The average drop in similarity after blurring the top 10, 20, 30 and 50 % of the patches (5, 10, 15 and 25 patches), the same cut-offs as for text.
- **Pointing game** ($\uparrow$). For each image we check whether the center of the highest-ranked patch lies inside a bounding box of the image’s main object. The accuracy is the share of images with a hit. As in the original definition [12], a tolerance of 15 pixels is allowed; we also report the strict result without tolerance.

The images have no bounding boxes, so we annotated one target object per image ourselves (Appendix C). The boxes were drafted by looking only at the images, never at the SHAP maps, and were then reviewed and approved. They are therefore *author annotations*, not benchmark ground truth. For scene images such as the crowd at a pool (I2) or the market stall (I7) we used one box around the group of people. Since the explained score is image-only, the pointing game here tells us whether the most important patch lies on the main object, not whether CLIP localizes a named class.

3 Implementation workflow

Figure 1 summarizes the path from an input to the evaluated explanation. It is the same for all three models; only the features, the way hidden features are represented and the explained output change.

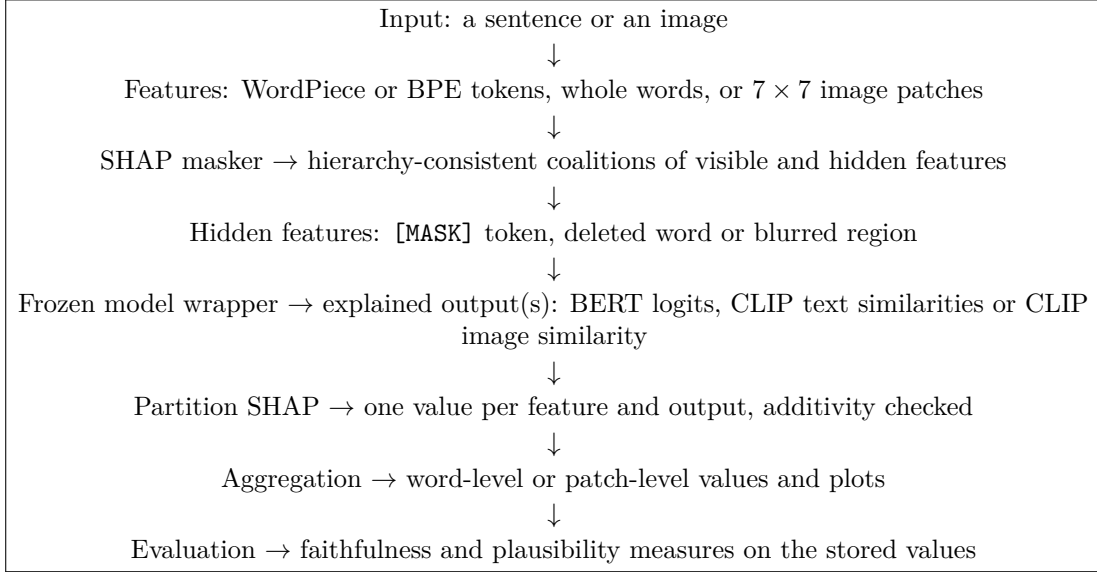


Figure 1: Processing sequence followed by the implementation.

Each experiment is a separate project with the same structure as in the previous report: a model wrapper that isolates tokenization or image preprocessing and inference, a module for masking and aggregation (WordPiece or BPE tokens to words, pixels to patches), a module for evaluation, a plotting module and a run script that handles configuration, input and output. The model wrappers evaluate every perturbed input without gradients and with the model in evaluation mode. Sub-word values are summed into words, and pixel values into 32×32 patches, only after SHAP has computed them; summation keeps Equation (1) valid.

The evaluation measures were implemented in two additional projects that read the stored SHAP values and never write into the original experiment folders. Because the measures were added to finished experiments, we built several checks into this code. Every run stops before writing any result if one of them fails:

- the stored SHAP values are complete and satisfy additivity;
- the rebuilt word or patch ranking and the rebuilt deletion inputs are identical to the ones stored by the original experiments;
- re-scoring the original inputs and the stored deletion inputs with the frozen models reproduces the stored scores (largest difference 1.3×10^{-5} logits for BERT and below 10^{-6} for CLIP);
- the recomputed deletion AOPC equals the value stored by the original 500-sentence experiment;
- the source folders are hashed before and after each run and must be unchanged.

4 Experimental setup

4.1 Data

Qualitative sentence set (S1–S10). The qualitative set of the previous report contains ten manually specified sentences, six positive and four negative (Appendix A). Several contain mixed

evidence, for example S3 (*slow* vs. *outstanding*) and S8 (*weak* vs. *captivating*). They are meant for inspection, not for measuring accuracy.

500 SST-2 sentences. For the quantitative text experiments we use a fixed, balanced subset of the SST-2 development set with 250 negative and 250 positive sentences. The subset was selected earlier in the project (excluding inspected and duplicate sentences) and was used unchanged; a SHA-256 hash of the file is checked before every run. Both text models explain exactly the same sentences in the same order.

Ten images (I1–I10). For the vision experiment we use ten natural photographs, for example a horse in a field, a kayak on the sea, two puppies and a child on a climbing net. CLIP resizes each image so that the shorter side is 224 pixels and keeps the central 224×224 crop; only this crop can receive attribution. The images come without any annotation.

4.2 Overview of the experiments

Table 1 summarizes the four experiments of this report. Experiments A and B are qualitative and explain the two class outputs separately. Experiment C explains the same 500 sentences with both text models and is the basis of the text measures. Experiment D explains the ten images.

Table 1: The four SHAP experiments of this report.

	Data	Explained output	Features and masking	Evaluation
A	S1–S10	BERT NEG and POS logits	WordPiece tokens replaced by [MASK]; values summed to words	additivity, qualitative, comparison with the previous report
B	S1–S10	CLIP text NEG and POS similarity	BPE tokens deleted (CLIP has no mask token); values summed to words	additivity, qualitative, agreement with A
C	500 SST-2	NEG, POS and POS–NEG margin for BERT and CLIP text	whole words deleted, remaining text re-tokenized by each model; identical for both	comprehensiveness, sufficiency, deletion AOPC, rationale P/R/F1
D	I1–I10	CLIP vision similarity to the original embedding	pixels replaced by a 128×128 box blur; values summed to 7×7 patches	deletion AUC, insertion AUC, AOPC, pointing game

Experiment A is the two-class version of the previous report and keeps its settings ([MASK] replacement without collapsing consecutive masks, maximum length 128), so that the new plots can be compared with the old ones; experiment B applies the same design to CLIP text. For experiment C we designed a shared *whole-word* masking: a sentence is split into words once (contractions such as *is n’t* and hyphenated words stay together, punctuation is a separate unit), a hidden word is deleted, and the remaining text is tokenized again by BERT’s WordPiece or CLIP’s BPE tokenizer. This way both models are explained over exactly the same features, and no word is ever removed only partially. Because A/B and C use different masking, their SHAP values should not be compared number by number.

The CLIP text prototypes also differ between the experiments. Experiment B uses five prompts per class (“This movie is terrible.”, “I hated this movie.”, “This is a bad film.”, “The film was disappointing.”, “This is a negative movie review.” and the positive counterparts). Experiment C uses a family of eight prompts per class that was selected earlier on a separate validation split of SST-2 training sentences, never on the 500 evaluation sentences (Appendix B). In experiment C each model explains three outputs at once: NEG, POS and the margin POS–NEG.

4.3 Configuration and reproducibility

Experiments A and B were run on CPU with Python 3.12, PyTorch 2.12, Transformers 5.9 and SHAP 0.52. Their runs also contained ten additional SST-2 sentences (D01–D10), which are not part of this report. Experiments C and D and all evaluation runs used one conda environment (Python 3.11, PyTorch 2.14 with CUDA 12.6, Transformers 5.4, SHAP 0.51) on an NVIDIA RTX 4060 Laptop GPU. Random seeds were fixed to 42, the SHAP budget was 500 model evaluations per sentence or image, and the batch size was 32.

On the GPU, Partition SHAP needed 0.37 s per sentence for BERT and 0.28 s for CLIP text (on average 300 model evaluations), and about one minute for all ten images. The text evaluation took about 20 seconds and the vision evaluation about 15 seconds. Running each evaluation twice produced byte-identical result files. No model was trained or fine-tuned in this project.

5 Results

5.1 Two-class explanations of BERT on S1–S10

BERT classifies all ten sentences correctly and with high confidence (margins between 6.9 and 8.4 logits, Table 2). For both classes and all ten sentences the base value plus the sum of the SHAP values reproduces the logit exactly (additivity residual 0 at the stored precision). The difference between the two class explanations reproduces the margin explanation of the previous report: for nine sentences the token values agree up to floating-point precision, and the largest difference is 0.023 in S3, which is explained by the finite evaluation budget.

Table 2: Predictions on S1–S10. The margin is POS–NEG (logits for BERT, cosine similarity for CLIP text).

ID	Gold	BERT		CLIP text	
		Prediction	Margin	Prediction	Margin
S1	POS	POS	8.246	POS	0.0125
S2	NEG	NEG	−7.147	NEG	−0.0023
S3	POS	POS	6.963	NEG	−0.0024
S4	NEG	NEG	−7.280	NEG	−0.0027
S5	POS	POS	8.167	POS	0.0068
S6	POS	POS	8.220	POS	0.0083
S7	NEG	NEG	−7.364	NEG	−0.0118
S8	POS	POS	6.937	POS	0.0072
S9	NEG	NEG	−7.151	NEG	−0.0054
S10	POS	POS	8.381	POS	0.0092
Correct		10 / 10		9 / 10	

Explaining the two classes separately shows a clear pattern: for BERT, almost every word moves the two logits in opposite directions by a similar amount. In S3 (Figure 2, top) *outstanding* lowers the NEG logit by 1.90 and raises the POS logit by 1.84, while *slow* raises NEG by 0.60 and lowers POS by 0.45. The difference between the two columns is the word’s contribution to the decision margin. Table 3 lists the word with the largest positive and the largest negative margin contribution in each sentence.

Table 3: Words with the largest positive (towards POS) and negative (towards NEG) contribution to the POS–NEG margin in each sentence.

ID	BERT (logits)		CLIP text (cosine)	
	towards POS	towards NEG	towards POS	towards NEG
S1	beautiful (+2.26)	of (−0.18)	film (+0.0092)	a (−0.0004)
S2	and (+2.34)	waste (−5.69)	movie (+0.0093)	waste (−0.0077)
S3	outstanding (+3.75)	slow (−1.04)	outstanding (+0.0026)	slow (−0.0012)
S4	a (+0.32)	dull (−4.01)	forgettable (+0.0016)	dull (−0.0020)
S5	brilliant (+4.05)	the (−0.51)	inspired (+0.0047)	The (−0.0010)
S6	breathhtaking (+1.85)	emotion (−1.72)	storytelling (+0.0052)	emotion (−0.0007)
S7	charm (+1.55)	boring (−3.57)	charm (+0.0035)	Painfully (−0.0046)
S8	captivating (+4.77)	weak (−3.88)	captivating (+0.0078)	weak (−0.0031)
S9	a (+0.50)	hollow (−3.69)	sequel (+0.0042)	disappointing (−0.0039)
S10	entertaining (+2.84)	to (−0.40)	entertaining (+0.0054)	end (−0.0003)

For BERT, the strongest word in the direction of the prediction is a clear sentiment word in every sentence: *waste*, *dull*, *boring* and *hollow* for the negative sentences, and *beautiful*, *outstanding*, *brilliant*, *captivating* and *entertaining* for the positive ones. This confirms the finding of the previous report. The contrastive sentences are handled sensibly as well: in S8 *captivating* (+4.77) and *weak* (−3.88) receive the two largest values with opposite signs, and the positive word wins, which matches the gold label.

Some values are less intuitive, as already noted in the previous report. In S2 the function word *and* receives +2.34, in S6 *emotion* pushes towards NEG although the sentence is positive, and in S7 the word *charm* pushes towards POS, although it appears in “devoid of any originality or charm”. SHAP values describe how the model reacts when words are hidden in the context of the other words; a word such as *charm* is positive on its own, and the negation lies in other words (*devoid*). These values should therefore be read as the model’s local behavior, not as a sentiment dictionary. All five plots of every sentence are given in Appendix F.

5.2 Two-class explanations of CLIP text on S1–S10

CLIP text classifies nine of the ten sentences correctly; it fails on the contrastive sentence S3 (Table 2). All explanations are again exactly additive. The most striking difference to BERT is the size of the decisions: CLIP’s margins lie between 0.002 and 0.012 cosine units, while both class scores are around 0.90–0.98. In other words, CLIP finds every sentence almost equally similar to the positive and the negative prototype, and the decision is made by a very small difference.

This is also visible in the explanations (Figure 2, bottom). Unlike BERT, CLIP’s NEG and POS values mostly have the *same* sign. Words that also occur in the prompts, such as *movie* in S2 or *film* in S1, raise the similarity to *both* prototypes (for S2, *movie* adds 0.050 to NEG and 0.059 to POS). They make the sentence look like a movie review, but they do not decide its sentiment. Only the margin column removes this shared part. After this correction the picture becomes more familiar (Table 3): *waste*, *dull*, *slow*, *weak* and *disappointing* push towards NEG, and *outstanding*, *captivating*, *charm* and *entertaining* towards POS. In S3 CLIP does see *outstanding* as positive and *slow* as negative, but the remaining words tip the tiny margin to NEG.

To compare the two models directly we correlated their word contributions to the margin. Over S1–S10, the mean Spearman rank correlation is 0.37, on average 1.8 of the three most important words are shared (Jaccard 0.46), and 59% of the words receive the same sign. The two models therefore agree on the most important sentiment words, but they rank the remaining words quite differently. All plots are given in Appendix G.

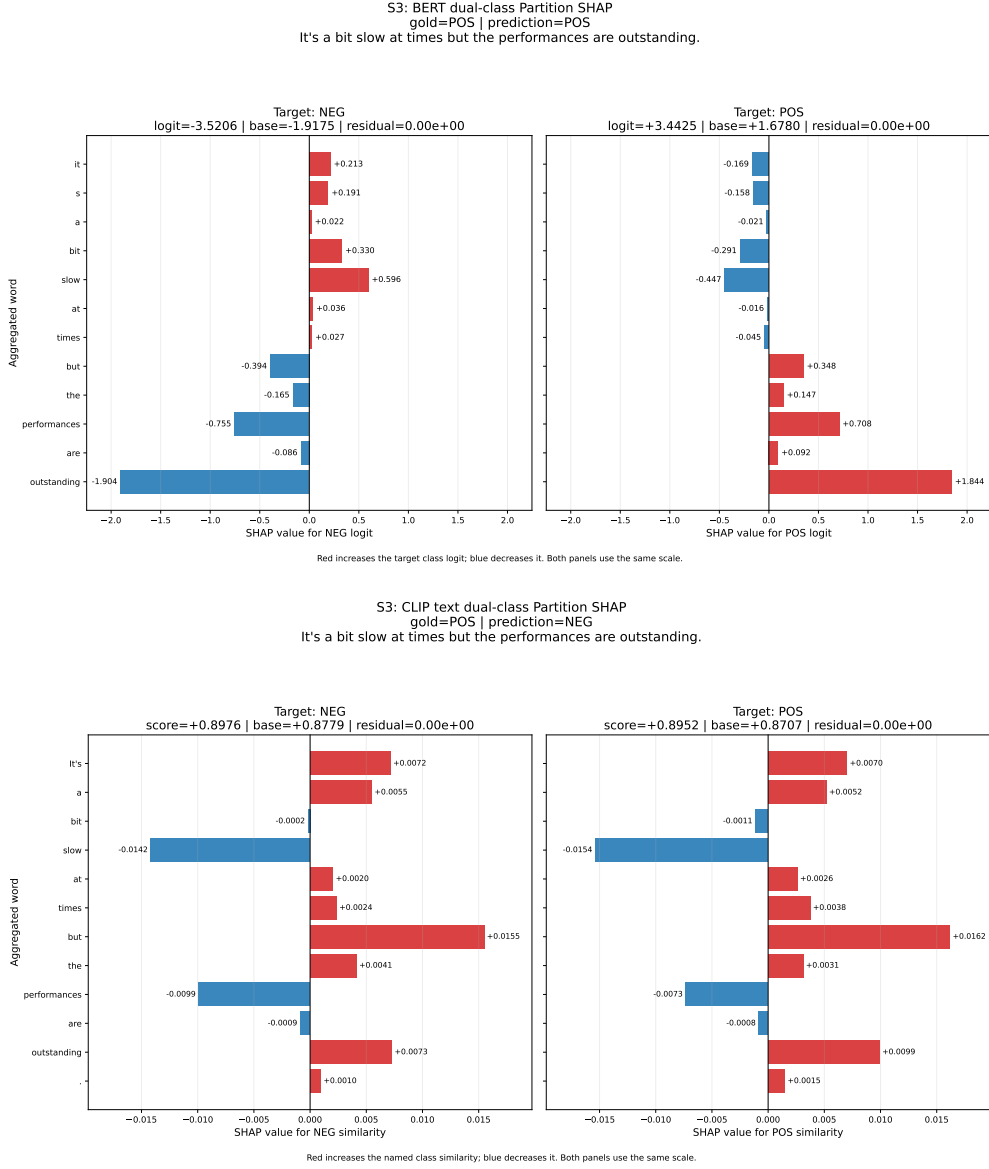


Figure 2: Sentence S3, “It’s a bit slow at times but the performances are outstanding.” Top: BERT (correct, POS). Bottom: CLIP text (wrong, NEG). Each panel shows the SHAP values of one class output; red increases and blue decreases the named class.

5.3 Faithfulness and plausibility on 500 sentences

Before looking at the explanations, it helps to know how good the two classifiers are on the 500 sentences (Table 4). BERT, which was trained for this task, reaches an accuracy of 93.6%. CLIP text, used zero-shot through prompts, reaches 70.2%. The SHAP explanations of both models passed the additivity check for all 500 sentences (largest residual 9.9×10^{-6} logits for BERT and 5.7×10^{-7} for CLIP).

Table 4: Classification results on the 500 SST-2 sentences (250 NEG, 250 POS).

Model	Accuracy	Macro-F1	NEG recall	POS recall
BERT (fine-tuned on SST-2)	0.936	0.936	0.924	0.948
CLIP text (zero-shot)	0.702	0.702	0.688	0.716

Table 5 shows the four explanation measures. BERT’s explanations are better on every measure.

Table 5: Explanation measures for the POS–NEG margin SHAP values on the 500 sentences. Rationale measures use the SST word ratings as a proxy reference and are computed on the 468 sentences that contain at least one sentiment word.

Metric	BERT	CLIP
Comprehensiveness $\uparrow$	0.7715	0.7049
Sufficiency $\downarrow$	0.2231	0.3307
Deletion AOPC $\uparrow$	0.8902	−0.3681
Rationale precision (SST proxy) $\uparrow$	0.4658	0.3277
Rationale recall (SST proxy) $\uparrow$	0.6778	0.4886
Rationale F1 (SST proxy) $\uparrow$	0.4935	0.3497

Comprehensiveness and sufficiency. Deleting the top-ranked words removes on average 77 % of a typical decision margin for BERT and 70 % for CLIP. Keeping only those words loses 22 % of a typical margin for BERT and 33 % for CLIP. For both models the explanations are clearly informative: a few words carry most of the decision. BERT’s explanations are somewhat more comprehensive and considerably more sufficient, i.e. BERT’s decision can be recovered better from the words SHAP selects. This extends the observation of the previous report from ten sentences to 500. Table 6 shows that both measures behave as expected with the size of the selection: comprehensiveness increases and sufficiency decreases the more words are selected.

Table 6: Text measures at each cut-off (means over the 500 sentences; rationale measures over 468 sentences). Mean k is the average number of selected words; the normalized deletion drop is the per-cut-off term of the deletion AOPC.

Cut-off	Mean k	Comprehensiveness $\uparrow$		Sufficiency $\downarrow$		Norm. deletion drop $\uparrow$	
		BERT	CLIP	BERT	CLIP	BERT	CLIP
10 %	2.1	0.560	0.453	0.285	0.421	0.614	−1.693
20 %	3.8	0.724	0.640	0.270	0.400	0.888	−0.683
30 %	5.5	0.846	0.779	0.209	0.302	0.995	0.238
50 %	8.6	0.956	0.948	0.128	0.200	1.064	0.665

Cut-off	Rationale precision $\uparrow$		Rationale recall $\uparrow$		Rationale F1 $\uparrow$	
	BERT	CLIP	BERT	CLIP	BERT	CLIP
10 %	0.604	0.381	0.460	0.284	0.490	0.305
20 %	0.508	0.348	0.632	0.421	0.528	0.356
30 %	0.426	0.314	0.753	0.545	0.511	0.374
50 %	0.326	0.267	0.866	0.704	0.445	0.364

Deletion AOPC. For BERT, deleting the top-ranked words removes on average 89 % of each sentence’s own margin. At 50 % of the words the average drop is even slightly larger than the margin itself (1.06), and 46 % of BERT’s predictions flip to the other class. For CLIP the mean is *negative* (−0.368). This does not mean that CLIP’s explanations point the wrong way. The per-sentence normalization divides by the sentence’s own margin, and for the 73 CLIP sentences with a margin below 0.001 a tiny change in similarity becomes a very large normalized value, in either direction. These few sentences dominate the mean. The median deletion AOPC, which is not affected by them, is 0.70 for CLIP and 0.67 for BERT. This is the reason why we normalized comprehensiveness and sufficiency by the model’s average margin instead: they use the same deletions as AOPC but are not dominated by near-zero margins.

Rationale F1. BERT’s selected words agree clearly better with the human sentiment ratings: 47 % of its selected words are sentiment words (CLIP 33 %), and it finds 68 % of the sentiment words (CLIP 49 %). As expected, precision falls and recall rises when more words are selected; the F1 score is highest at 20–30 % of the words for both models. Two properties of the proxy reference should be kept in mind. First, it does not capture sentiment that arises from context: in “the film tries too hard to be funny”, *funny* counts as a sentiment word while *tries too hard*, which carries the negative judgment, does not. Second, a few mildly rated words such as *out*, *us* or *life* fall just outside the neutral band and count as sentiment words. The absolute F1 values are therefore not very meaningful on their own, but the comparison between the two models is fair, because both are measured against the identical reference.

5.4 Patch-level explanations of CLIP vision on I1–I10

Figure 3 shows two of the ten patch explanations; all ten are given in Appendix H. In all images every patch received a non-negative value, i.e. every visible region helps CLIP to keep its representation of the image. The values are small (the largest patch value per image is between 0.020 and 0.053, while the gap between the blurred and the original image is about 0.5), which shows that the representation is spread over many patches. The largest values lie on the main object in most images: on the kayak and paddler in I4, on the puppy’s head in I5, on the child’s face in I10 and on the man’s cap in I8. In I1, however, the highest value lies on the burning stump in the lower right corner and not on the horse, which is the largest object in the picture. A repeated run of I1 with 1,000 instead of 500 SHAP evaluations gave almost the same patch values (Spearman correlation 0.98, same top patch), so these maps are stable with respect to the evaluation budget.

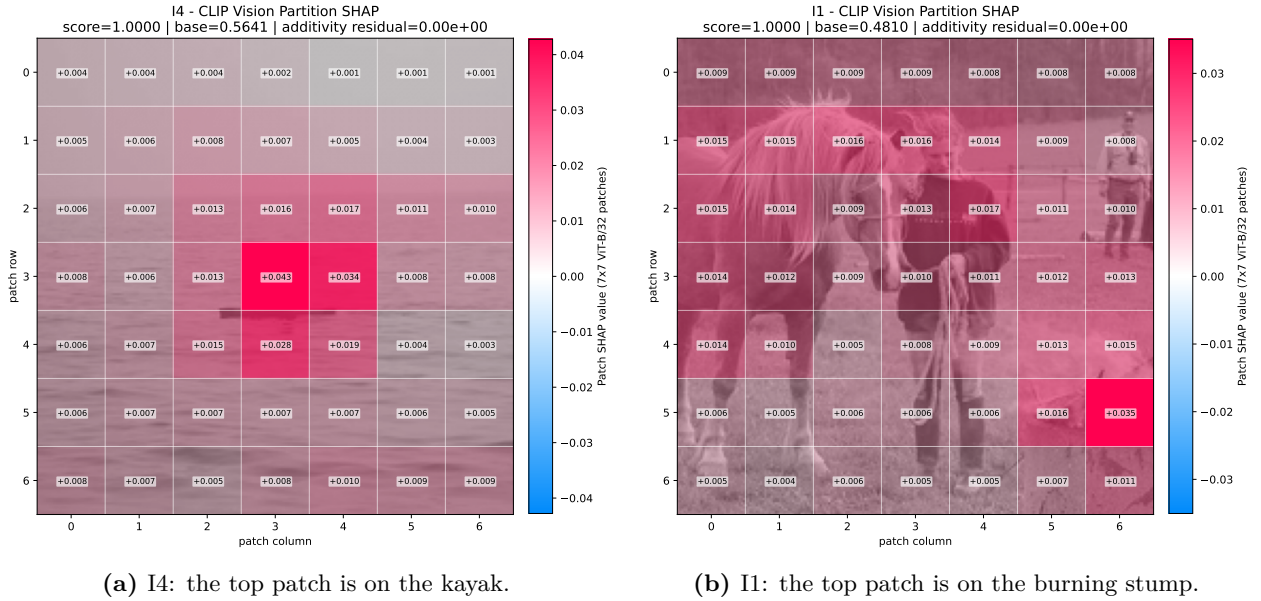


Figure 3: Patch-level SHAP values of CLIP vision for two images. Each cell is one 32×32 ViT patch of the 224×224 input; darker red means a larger contribution to keeping the image embedding.

5.5 Faithfulness and pointing game for CLIP vision

Table 7 summarizes the four image measures and Table 8 gives the values per image.

Table 7: Explanation measures for CLIP vision (means over I1–I10).

Metric	CLIP Vision
Deletion AUC ↓	0.7109
Insertion AUC ↑	0.8492
AOPC ↑	0.1995
Pointing/localization accuracy ↑ without pixel tolerance	0.9000 0.7000

Table 8: CLIP vision measures per image. “Blurred” is the similarity of the fully blurred image, i.e. the end of the deletion curve and the start of the insertion curve.

Image	Blurred	DAUC ↓	IAUC ↑	AOPC ↑	Target object	Hit (15 px)	Hit (0 px)
I1	0.481	0.742	0.878	0.182	horse	no	no
I2	0.507	0.734	0.875	0.134	people	yes	no
I3	0.471	0.768	0.890	0.156	person	yes	yes
I4	0.564	0.768	0.933	0.179	kayak with paddler	yes	yes
I5	0.499	0.656	0.882	0.336	dogs	yes	yes
I6	0.296	0.637	0.764	0.208	puppet-theatre cart	yes	yes
I7	0.445	0.686	0.806	0.193	people (vendors)	yes	yes
I8	0.430	0.678	0.802	0.218	person	yes	yes
I9	0.408	0.765	0.830	0.156	people (couple)	yes	no
I10	0.420	0.676	0.832	0.233	child	yes	yes

Deletion and insertion. Figure 4 shows the curves. Blurring the single most important patch already lowers the mean similarity from 1 to about 0.92; after one quarter of the patches it is about 0.82 and after half of them about 0.72. In the insertion direction, restoring the single most important patch raises the similarity of the blurred image from 0.45 to 0.56, and restoring the first quarter of the patches brings it to about 0.78. Both curves therefore move fastest at the beginning, which is what we expect when the ranking puts the important patches first. The deletion AUC of 0.71 must be read together with the baseline: the deletion curve cannot fall below the similarity of the fully blurred image (0.45 on average), so an area close to zero is impossible here. The same holds for AOPC: blurring the top 10–50 % of the patches lowers the similarity by 0.20 on average, compared with a maximum possible drop of about 0.55.

Pointing game. In 9 of the 10 images the center of the highest-ranked patch lies on the annotated object (Figure 5). Two of these hits (I2 and I9) are only within the 15-pixel tolerance: the point lies 10 and 12 pixels outside the box. Without tolerance the accuracy is therefore 0.70. The clear miss is I1, where the top patch lies on the fire. We should also note that the group boxes in I2 and I7 are large, which makes a hit easier than for a small single object such as the fisherman in I3 or the couple in I9.

6 Discussion

What the results tell us. Across all experiments Partition SHAP produced explanations that are internally consistent: every explanation reproduces the explained model output, and the evaluation reproduced all stored scores. For BERT the explanations are also convincing from the outside. The two-class explanations agree with the margin explanation of the previous report and show that each word acts on both logits in opposite directions (objective 1). On 500 sentences, removing the

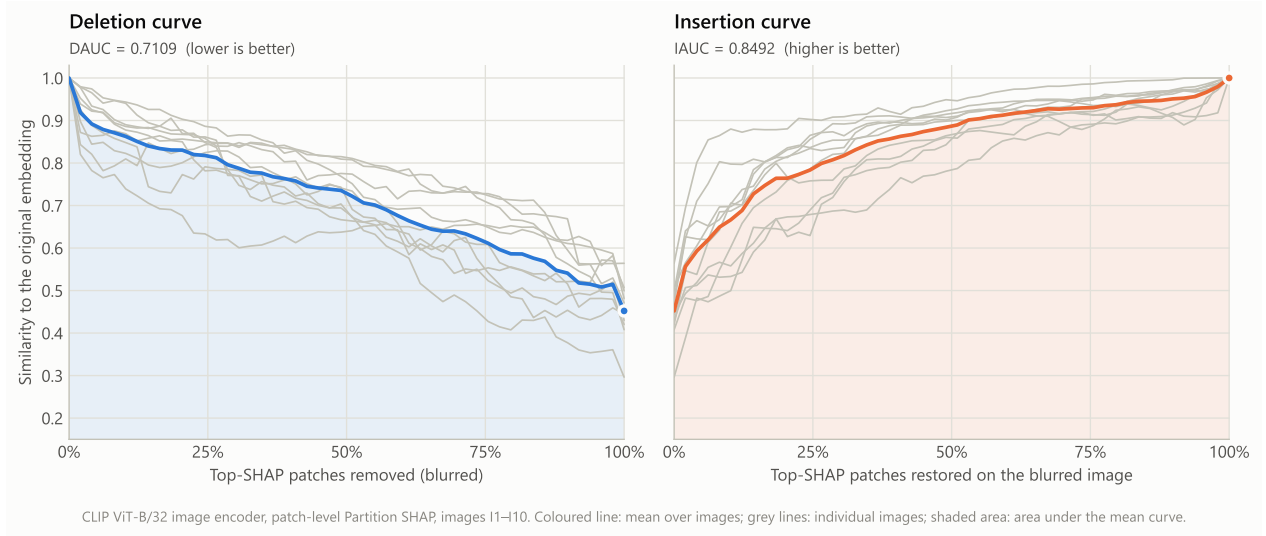


Figure 4: Deletion (left) and insertion (right) curves of CLIP vision. Gray: individual images; colored: mean over the ten images; shaded: area under the mean curve. The per-image curves are shown in Appendix D.

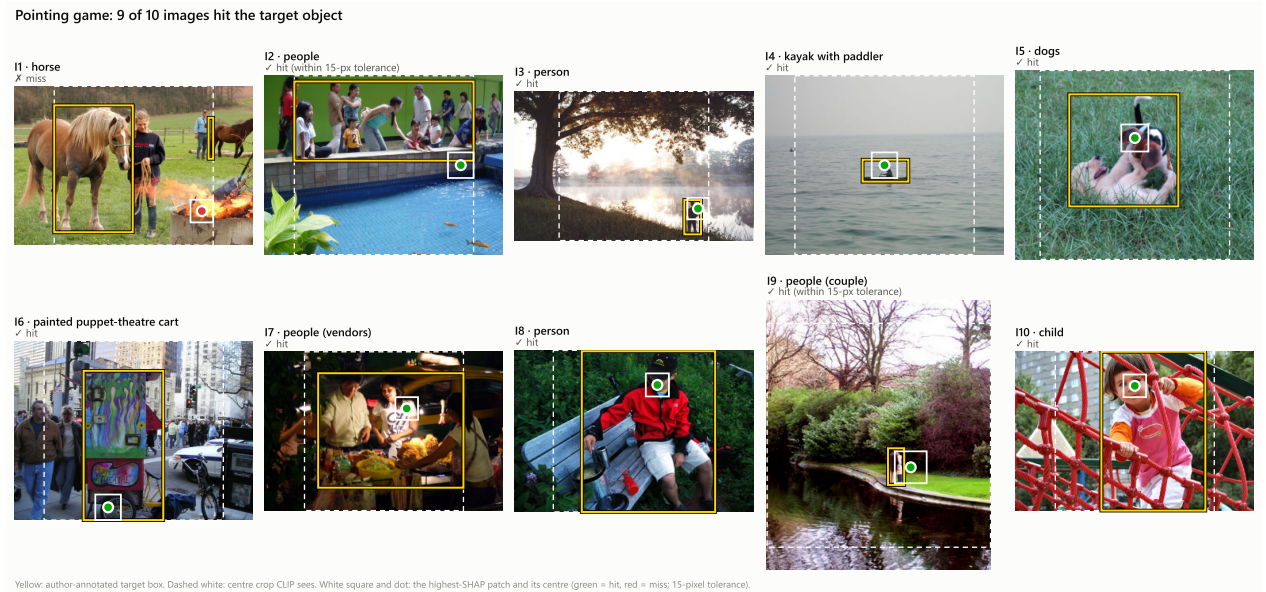


Figure 5: Pointing game. Yellow: annotated target box; dashed white: the center crop CLIP sees; white square and dot: the highest-ranked patch and its center (green = hit, red = miss).

top-ranked words destroys most of the decision while keeping them preserves most of it (objective 3), and the selected words agree reasonably well with human word ratings (objective 4). The main finding of the previous report therefore holds on a much larger set.

BERT versus CLIP text. The explanations of CLIP text are weaker on every measure (objectives 2–4), but we think this mostly reflects the model and not the explanation method. CLIP was never trained for sentiment; with prompts it reaches 70 % accuracy, and its decisions rest on very small differences between two large similarities. In such a model, many words change both class scores together (for example words that appear in the prompts), and the part that decides the sentiment is small. SHAP faithfully reports this, which is why the CLIP explanations are less concentrated on sentiment words. The comparison also taught us that the choice of normalization matters: with the per-sentence normalization of the original AOPC, the mean for CLIP is negative and says little about the explanations, while the same deletions normalized by the model’s typical margin give a clear and comparable result.

CLIP vision. For images SHAP also produced faithful rankings (objective 5): the curves change fastest for the first patches and the most important patch usually lies on the main object. The explained score is, however, special. Because CLIP vision has no class output, we explain how well the image keeps its own embedding. The most important regions are then the regions that make the image recognizable as a whole, which often, but not always, is the main object. I1 is a good example: the fire is small but visually very distinctive, and CLIP apparently needs it to represent this particular image.

Limitations. Several choices limit how far the numbers can be interpreted.

- **Masking baseline.** All SHAP values and all deletion measures depend on what a “missing” word or patch is: a [MASK] token, a deleted word or a blurred region. The models never saw such inputs during training. We used the same definition for explanation and evaluation, which keeps the comparison consistent, but a different baseline could give different values.
- **Partition approximation.** Partition SHAP computes Owen values with at most 500 evaluations, not exact Shapley values. The stability check on I1 suggests that this budget is sufficient, but we checked it only for one image.
- **Proxy references.** Rationale F1 uses context-free word ratings instead of sentence-specific human rationales, and the pointing game uses boxes that we annotated ourselves. Both are reasonable substitutes, but they are not benchmark ground truth, and we report them as such.
- **Small qualitative sets.** S1–S10 and I1–I10 are ten examples each. They are useful to understand the explanations, but the vision measures in particular rest on ten images only.
- **Different settings across experiments.** Experiments A/B and C use different masking and different CLIP prompts, so their SHAP values are not directly comparable. We therefore only compare BERT and CLIP *within* one experiment.
- **No uncertainty estimates.** We report plain averages without confidence intervals, as required for this part of the project. Small differences, such as the comprehensiveness gap of 0.07, should therefore not be over-interpreted.

What we learned. Working on this project, we found that an explanation method is only half of the work; the other half is making sure the numbers mean what we think they mean. Checking additivity, re-scoring the stored inputs and hashing the source folders caught several subtle problems

during development, for example when a quantity we had quoted turned out to be a difference between two metrics. We also learned to be careful with normalization and with references that only approximate human judgment.

7 Conclusion

We extended our previous SHAP analysis of BERT to two-class explanations, to the text and vision encoders of CLIP, and to a quantitative evaluation with established measures. For BERT on SST-2 the explanations identify the expected sentiment words, are faithful (comprehensiveness 0.772, sufficiency 0.223, deletion AOPC 0.890) and agree with human word ratings (rationale F1 0.494 with the SST proxy reference). For CLIP text used as a zero-shot sentiment classifier, the explanations are less faithful and less plausible (comprehensiveness 0.705, sufficiency 0.331, rationale F1 0.350), which matches its weaker classification (70 % vs. 94 % accuracy) and its very small decision margins. For CLIP vision the patch rankings are faithful (deletion AUC 0.711, insertion AUC 0.849, AOPC 0.199) and the most important patch lies on the main object in 9 of 10 images.

Several extensions would make the results stronger: sentence-level human rationales (for example from the ERASER movie review data) instead of the lexical proxy, images with official bounding boxes (for example from COCO or PASCAL VOC), a class-specific target for CLIP vision such as the similarity to a caption, other masking baselines, and confidence intervals for all measures. A comparison with other attribution methods such as Integrated Gradients would show how much of our findings is specific to SHAP.

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A Qualitative sentence set

Table 9: Complete qualitative sentence set used in experiments A and B.

ID	Gold	Sentence
S1	POS	The film is a beautiful and moving portrait of human resilience.
S2	NEG	This movie is an absolute waste of time and money.
S3	POS	It’s a bit slow at times but the performances are outstanding.
S4	NEG	A dull, tedious and completely forgettable experience.
S5	POS	The direction is inspired and the acting is nothing short of brilliant.
S6	POS	A masterpiece of storytelling with breathtaking visuals and emotion.
S7	NEG	Painfully boring and utterly devoid of any originality or charm.
S8	POS	The screenplay is weak but the lead actor delivers a captivating turn.
S9	NEG	A hollow and disappointing sequel that betrays everything the original stood for.
S10	POS	Funny, heartfelt and endlessly entertaining from beginning to end.

B CLIP text prompt families

Table 10: Prompts used to build the CLIP text prototypes.

Experiment	NEG prompts	POS prompts
B (S1–S10)	This movie is terrible. / I hated this movie. / This is a bad film. / The film was disappointing. / This is a negative movie review.	This movie is excellent. / I loved this movie. / This is a good film. / The film was enjoyable. / This is a positive movie review.
C (500 sentences)	This is a negative review. / This movie is terrible. / The film was disappointing. / The reviewer disliked the film. / This film left a bad impression. / The overall judgement is negative. / This was an unenjoyable experience. / I would not recommend this film.	This is a positive review. / This movie is excellent. / The film was satisfying. / The reviewer liked the film. / This film left a good impression. / The overall judgement is positive. / This was an enjoyable experience. / I would recommend this film.

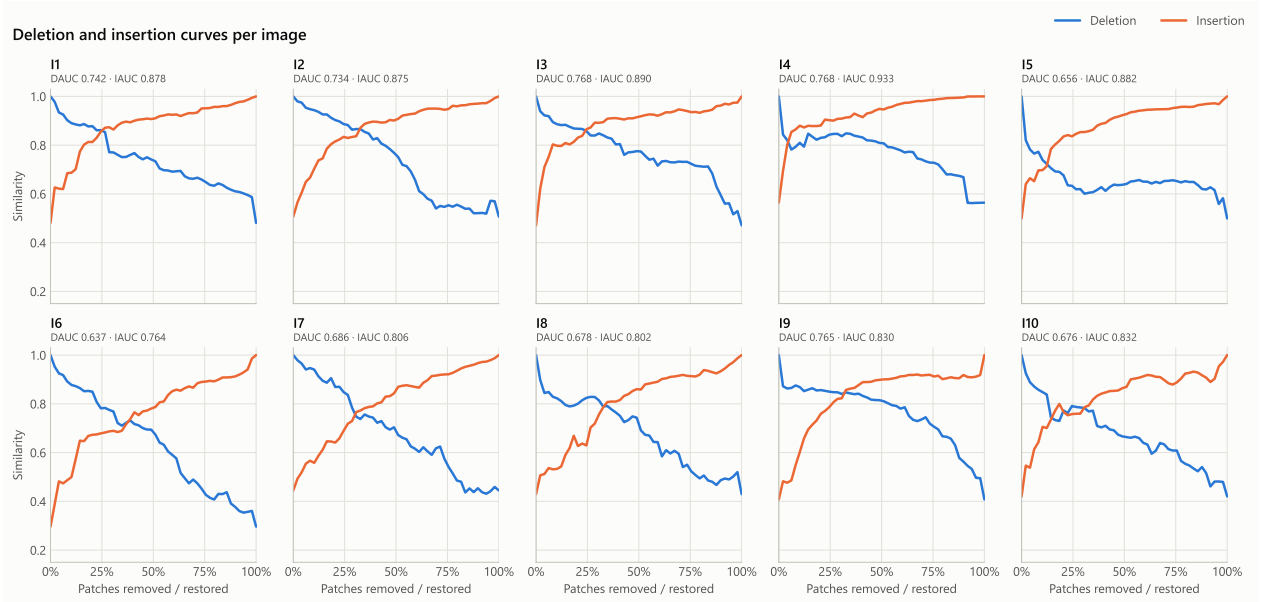
C Target boxes for the pointing game

Coordinates are pixels of the original image (x_0, y_0 = top left, x_1, y_1 = bottom right). The boxes were drafted without looking at the SHAP maps and approved on 30 September 2026; for the evaluation they are clipped to the center crop CLIP sees.

Table 11: Author-annotated target boxes.

Image	Target	x_0	y_0	x_1	y_1	Note
I1	horse	8	40	248	306	main horse
I1	horse	407	66	500	152	background horse, mostly outside the crop
I2	people	25	12	500	178	one box around the group at the pool edge
I3	person	355	228	390	300	man fishing; rest is landscape
I4	kayak with paddler	204	176	300	222	
I5	dogs	90	40	273	228	both puppies
I6	painted puppet-theatre cart	145	62	312	375	booth with wheels
I7	people (vendors)	112	45	482	285	three people at the stall
I8	person	140	2	420	339	man in the red jacket
I9	people (couple)	224	274	254	342	couple by the pond
I10	child	180	0	398	333	child on the rope net

D Additional vision results

**Figure 6:** Deletion and insertion curves for each image.

E Reproducibility record

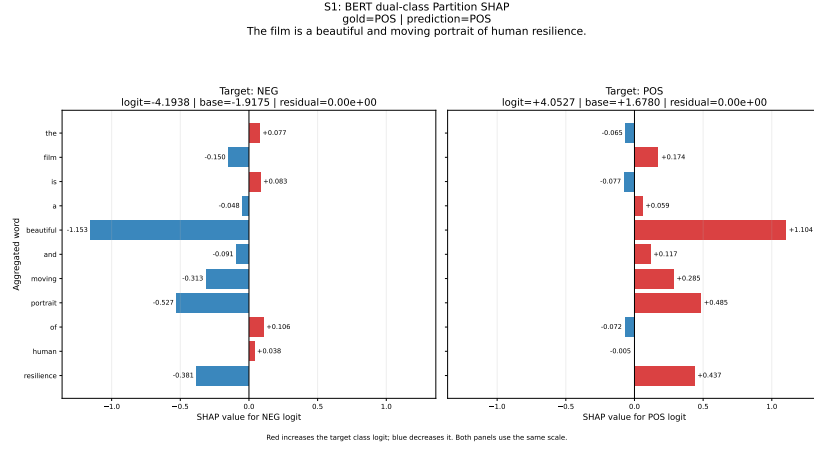
Table 12: Where the results of this report come from.

Part	Project folder	Report material
A	<code>bert_shap/dual_class_bert</code> (run <code>bert_dual_20_curated</code>)	<code>02_dual_class_bert_S1_S10</code>
B	<code>clip_text_shap</code> (run <code>clip_text_dual_20_curated</code>)	<code>03_clip_text_shap_S1_S10</code>
C	<code>bert_clip_partition_shap_500</code> (SHAP values); <code>bert_clip_shap_faithfulness_metrics</code> (measures)	<code>01_bert_clip_faithfulness_metrics</code>
D	<code>clip_vision_shap</code> (run <code>clip_vision_i1_i10</code>); <code>clip_vision_shap_faithfulness_metrics</code> (measures, boxes)	<code>04_clip_vision_shap_I1_I10_patch_overlay_values</code> , <code>05_clip_vision_faithfulness_metrics</code>

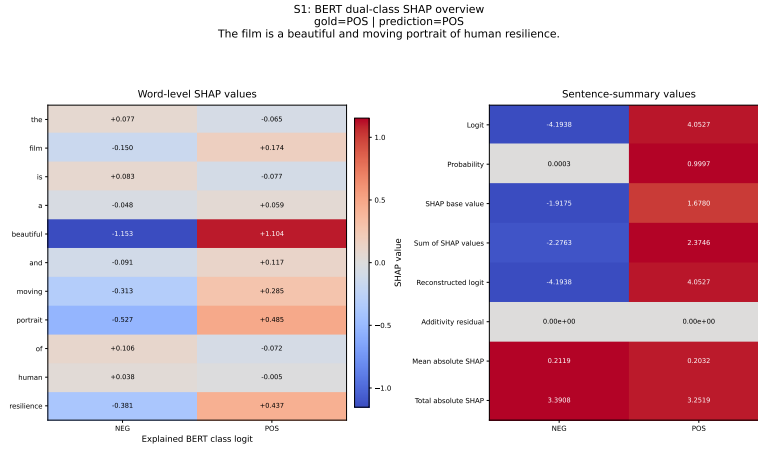
Settings shared by all SHAP runs: Partition explainer, at most 500 model evaluations per sentence or image, batch size 32, seed 42. BERT maximum length 128 tokens, CLIP text maximum length 77 tokens, CLIP vision input 224×224 with a 128×128 box blur. No model was trained or fine-tuned in this project. The evaluation projects contain a `README.md` with all definitions and checks, a `checks.json` with every verification result, and a batch file that re-runs the evaluation and its tests (23 tests for the text measures, 13 for the image measures).

F Complete BERT sentence-level plots

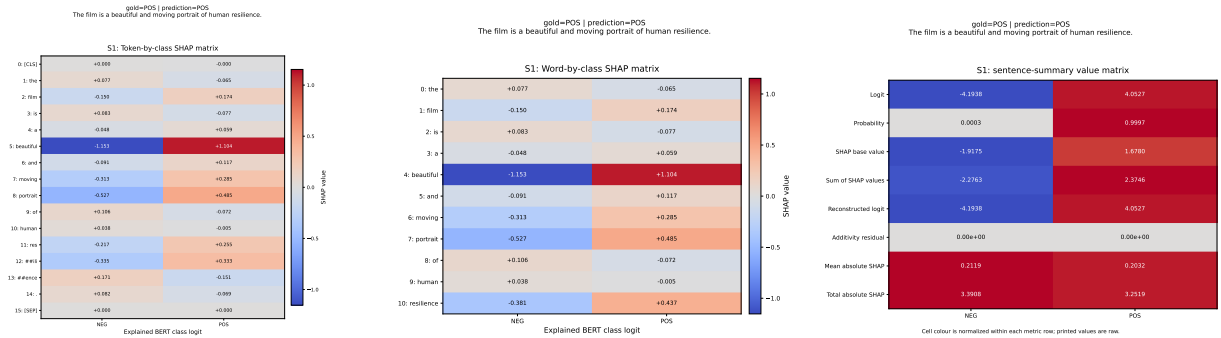
Each page shows the five plots of one sentence. In the word bars and matrices, red increases and blue decreases the named class logit. The summary matrix reports the logits, probabilities, base values, the sum of the SHAP values, the reconstructed logits and the additivity residual.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

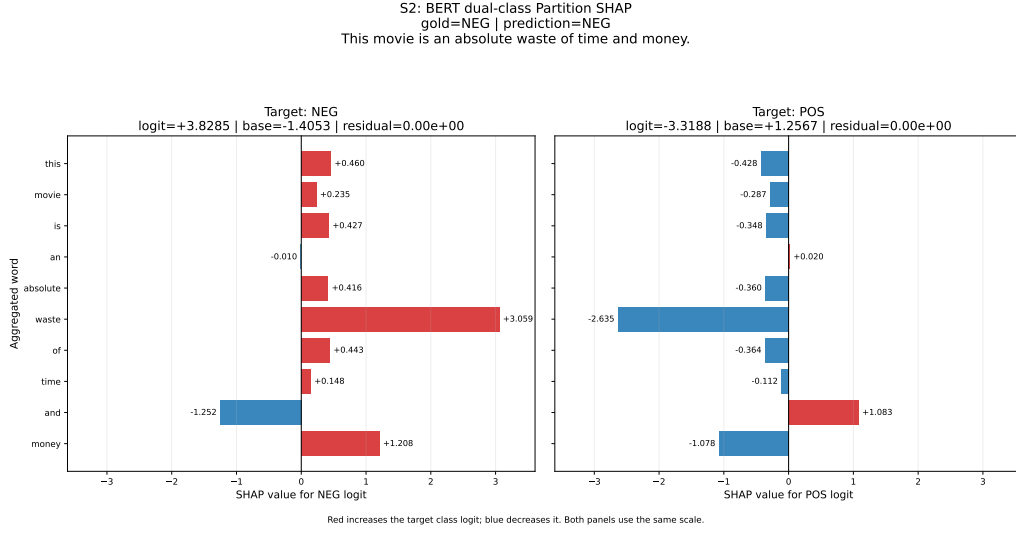


(c) Token-by-class matrix

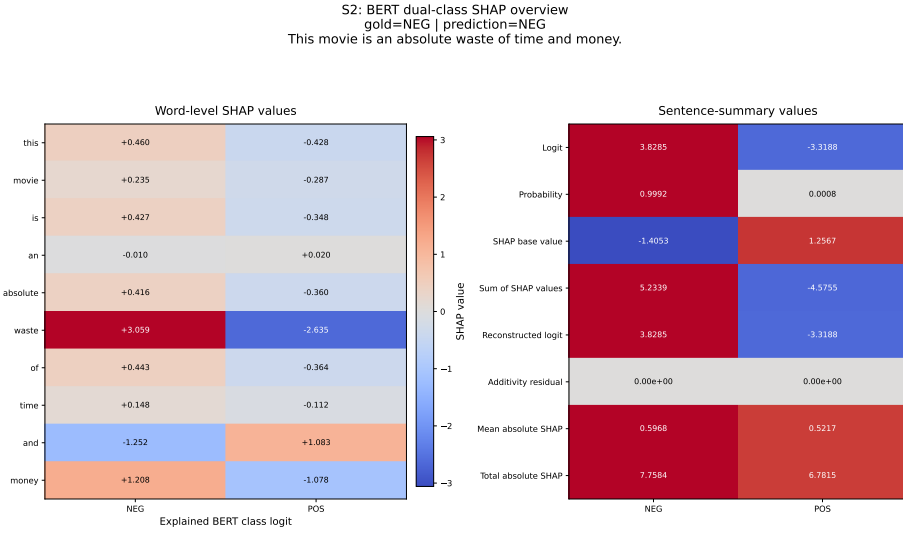
(d) Word-by-class matrix

(e) Sentence summary matrix

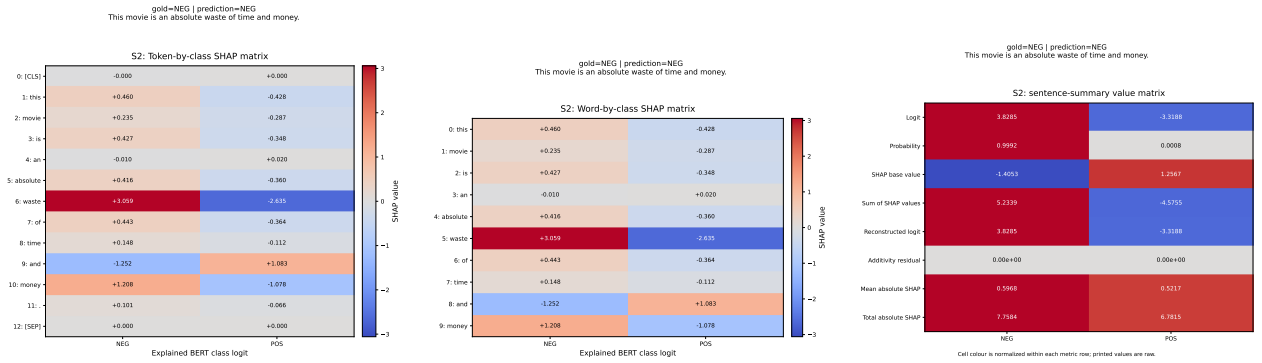
Figure 7: BERT, sentence S1: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

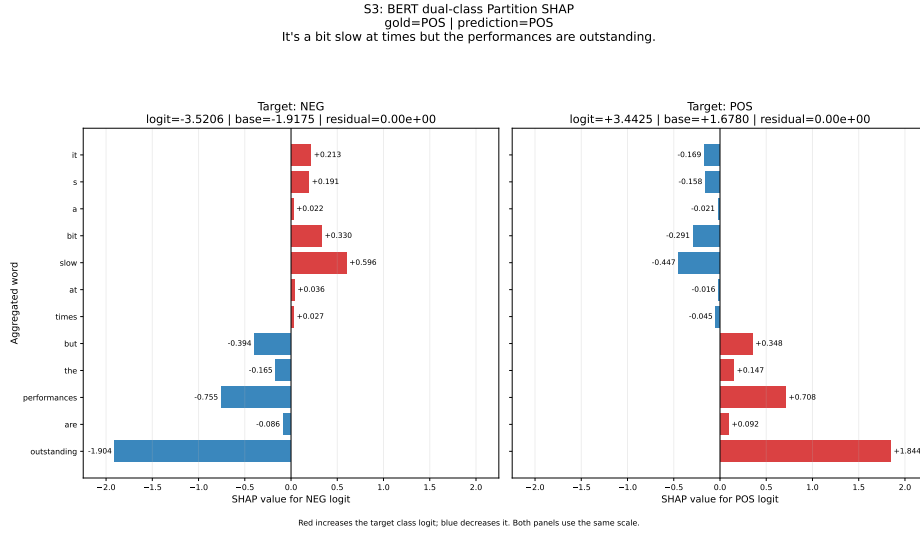


(c) Token-by-class matrix

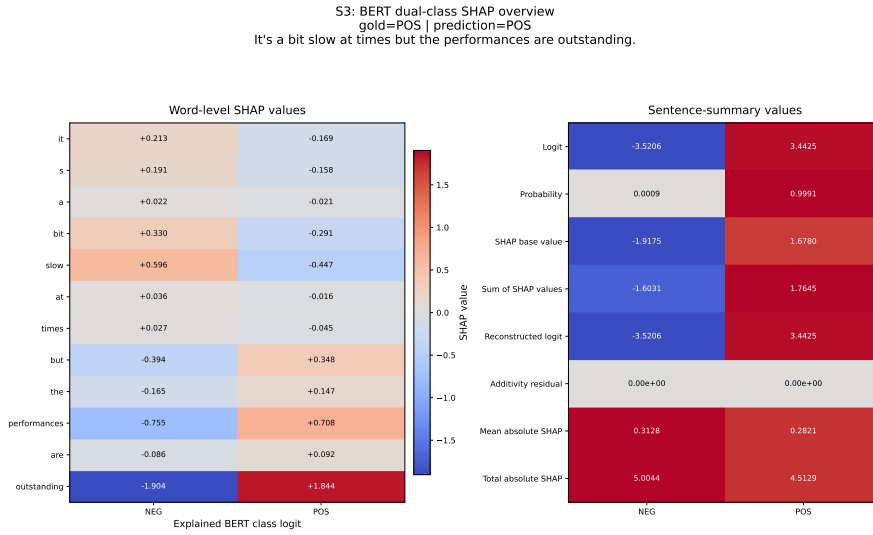
(d) Word-by-class matrix

(e) Sentence summary matrix

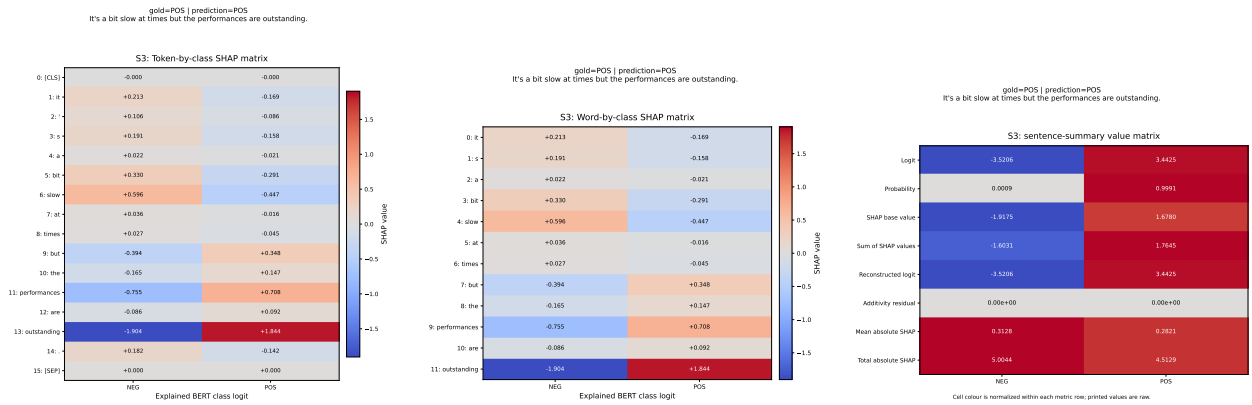
Figure 8: BERT, sentence S2: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

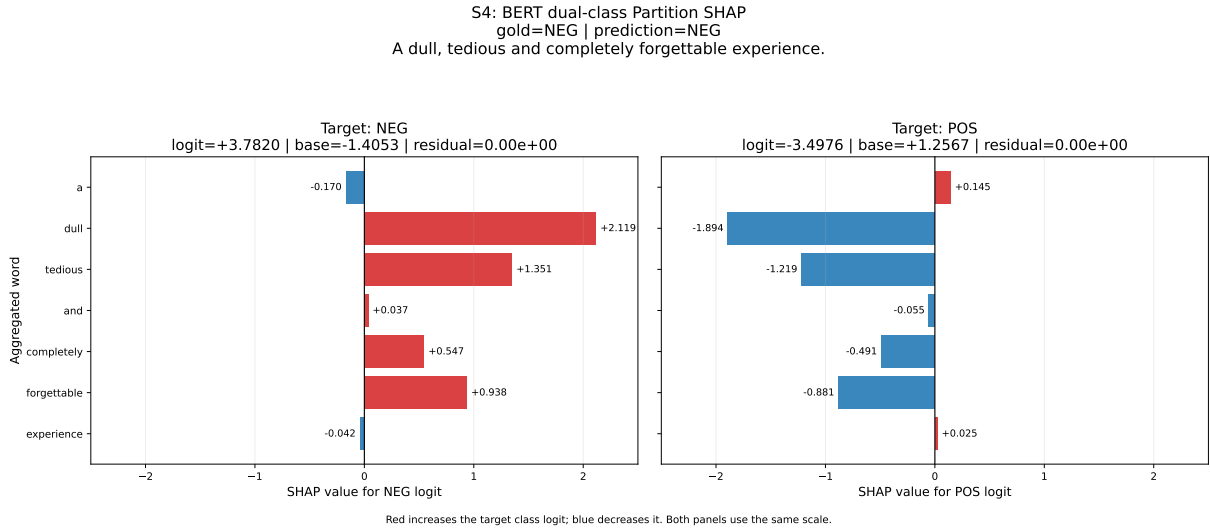


(c) Token-by-class matrix

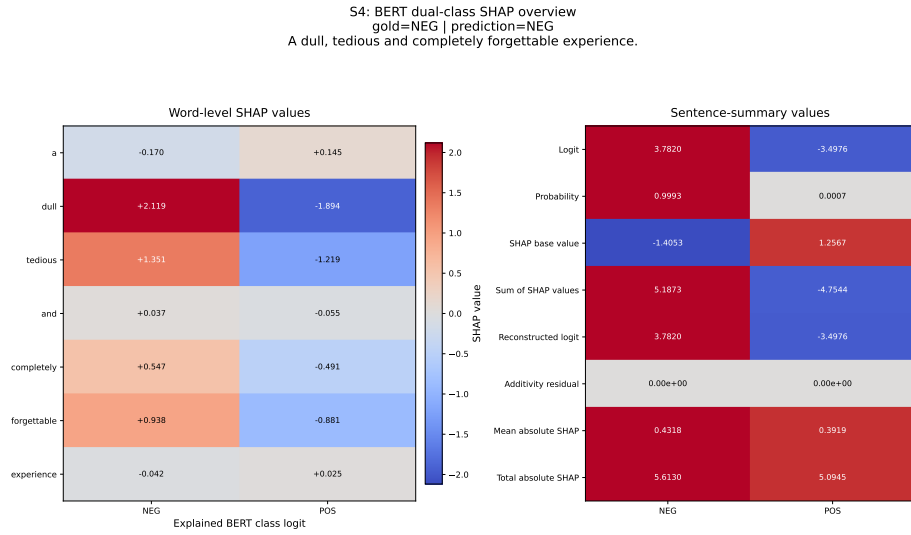
(d) Word-by-class matrix

(e) Sentence summary matrix

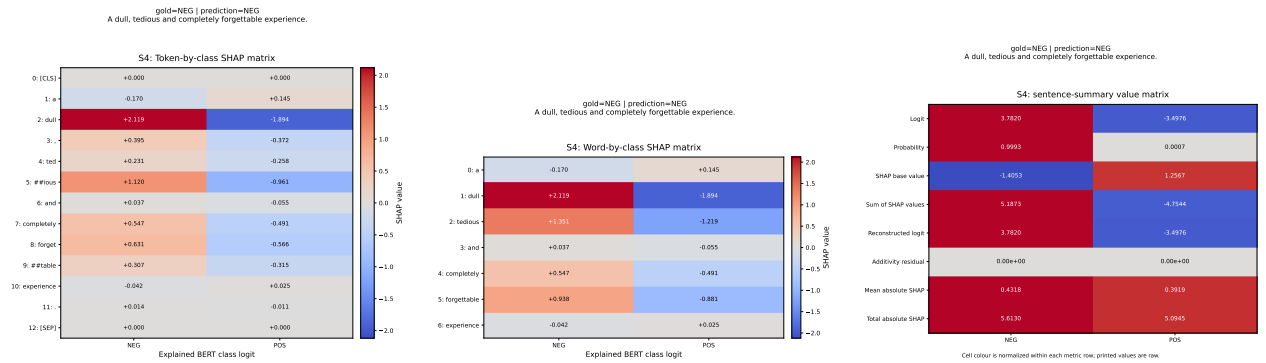
Figure 9: BERT, sentence S3: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

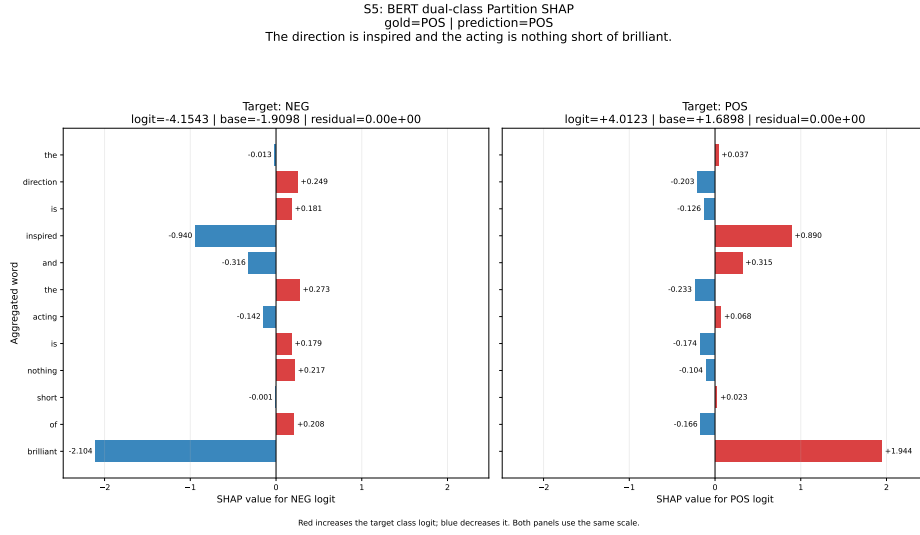


(c) Token-by-class matrix

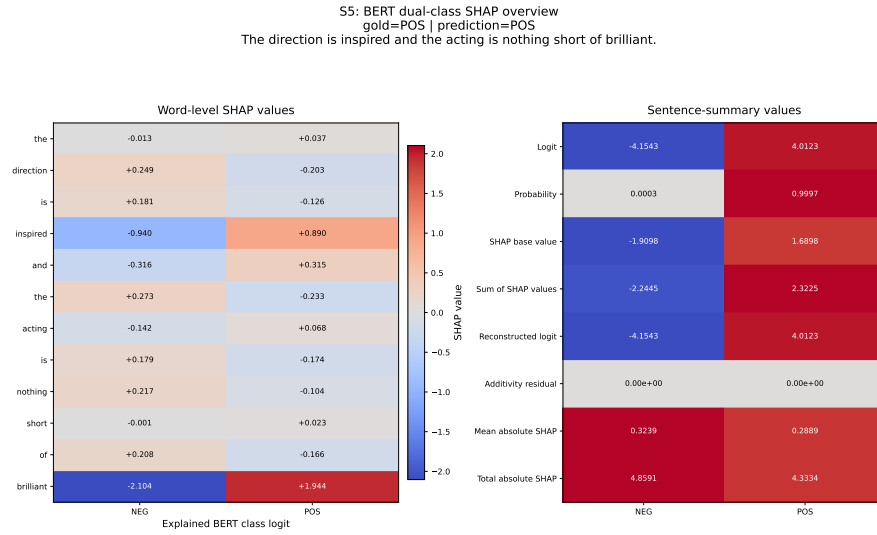
(d) Word-by-class matrix

(e) Sentence summary matrix

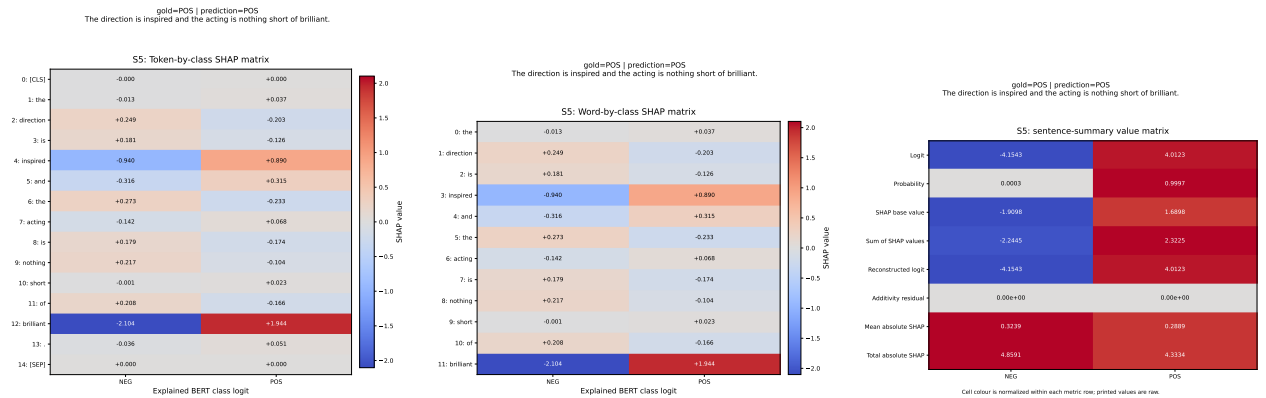
Figure 10: BERT, sentence S4: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

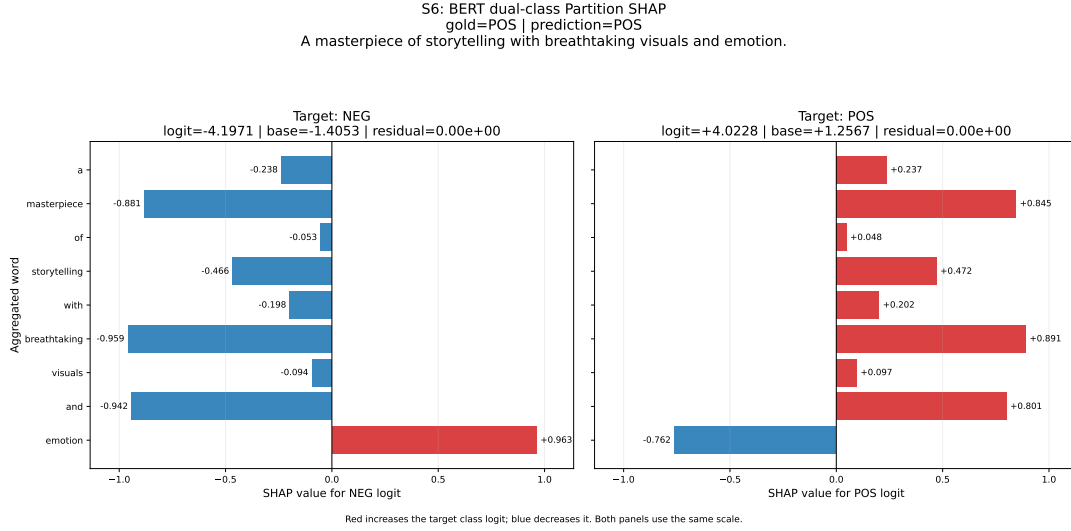


(c) Token-by-class matrix

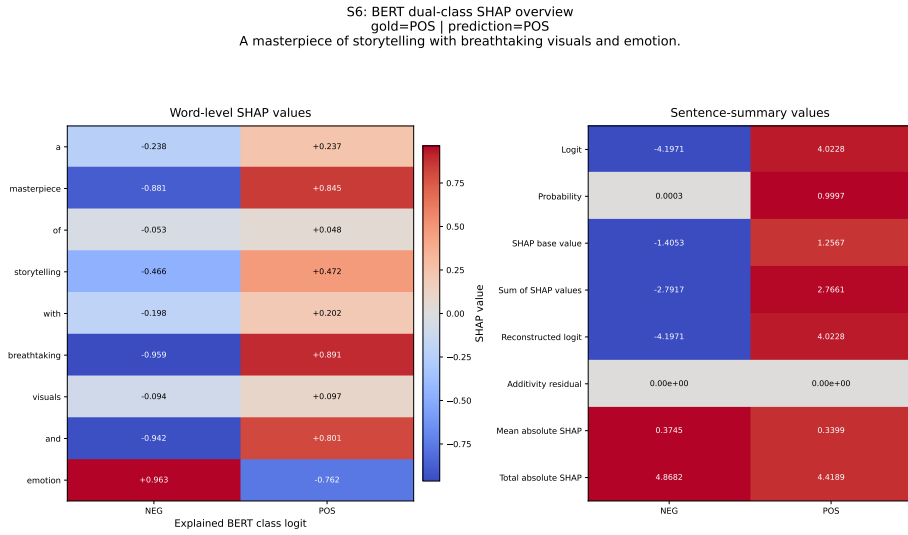
(d) Word-by-class matrix

(e) Sentence summary matrix

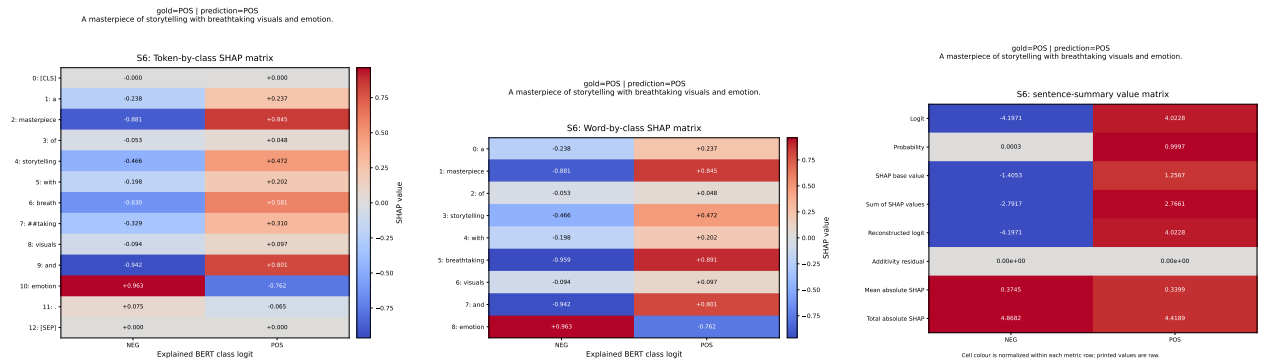
Figure 11: BERT, sentence S5: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

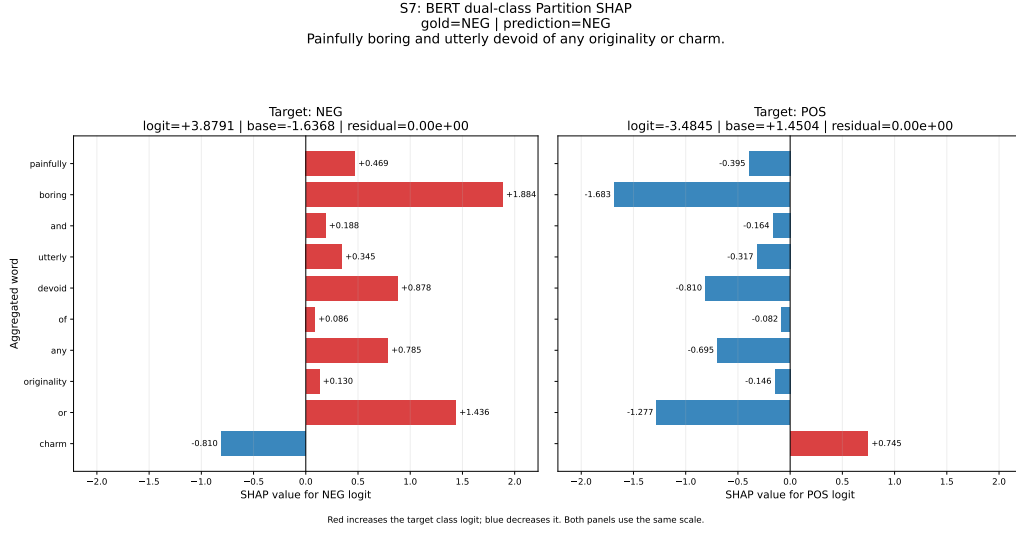


(c) Token-by-class matrix

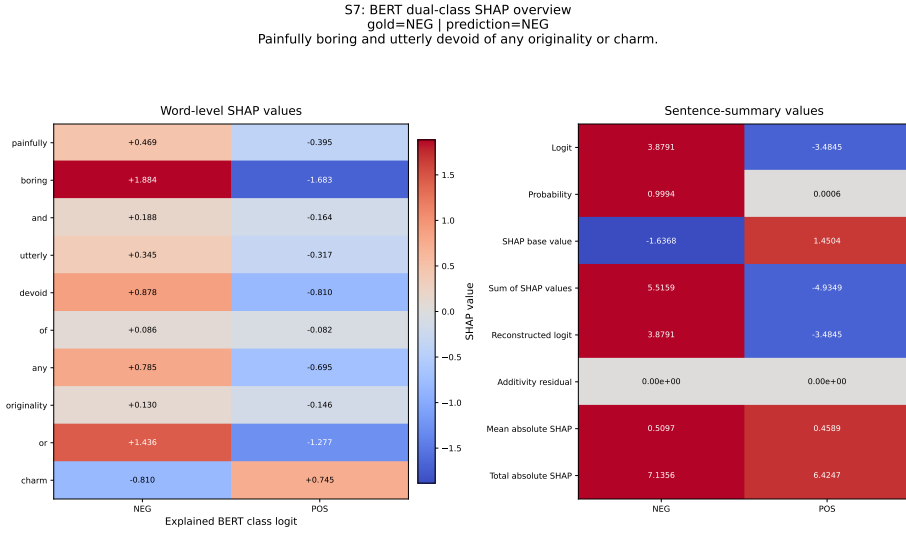
(d) Word-by-class matrix

(e) Sentence summary matrix

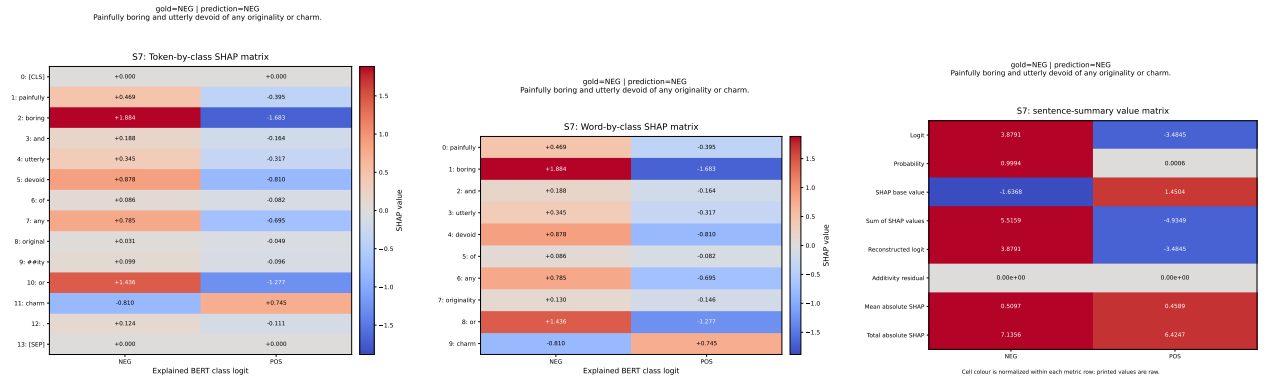
Figure 12: BERT, sentence S6: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

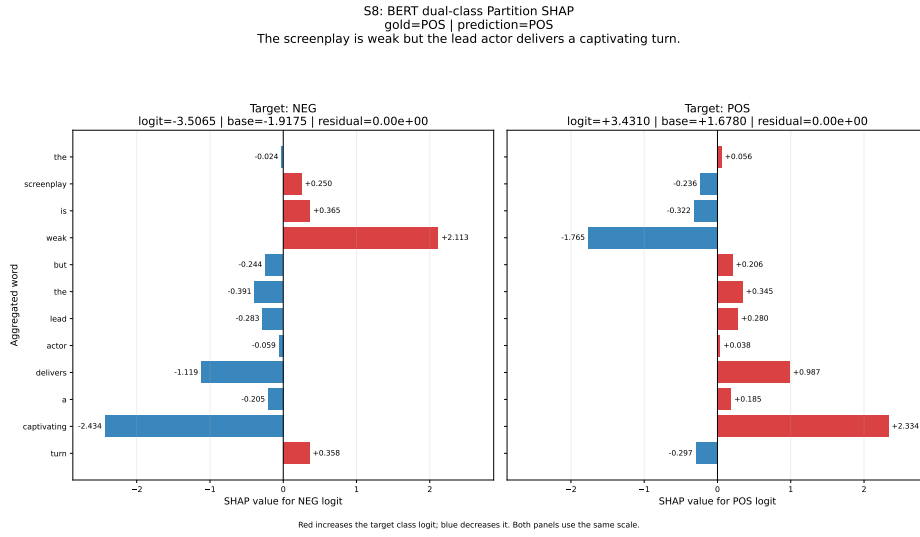


(c) Token-by-class matrix

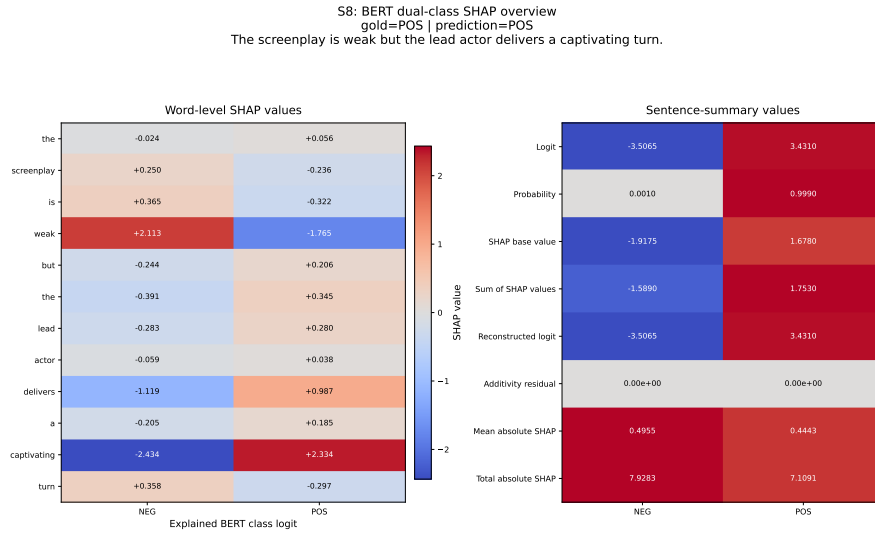
(d) Word-by-class matrix

(e) Sentence summary matrix

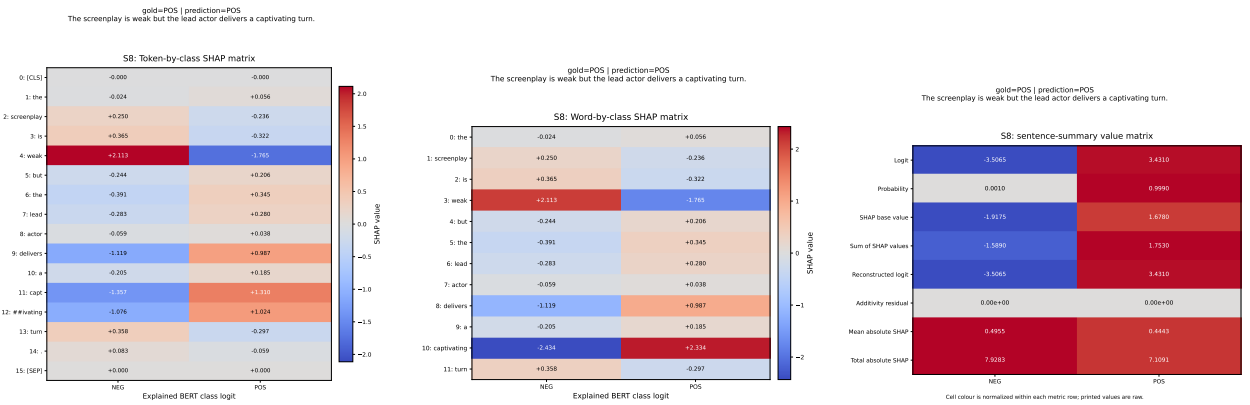
Figure 13: BERT, sentence S7: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

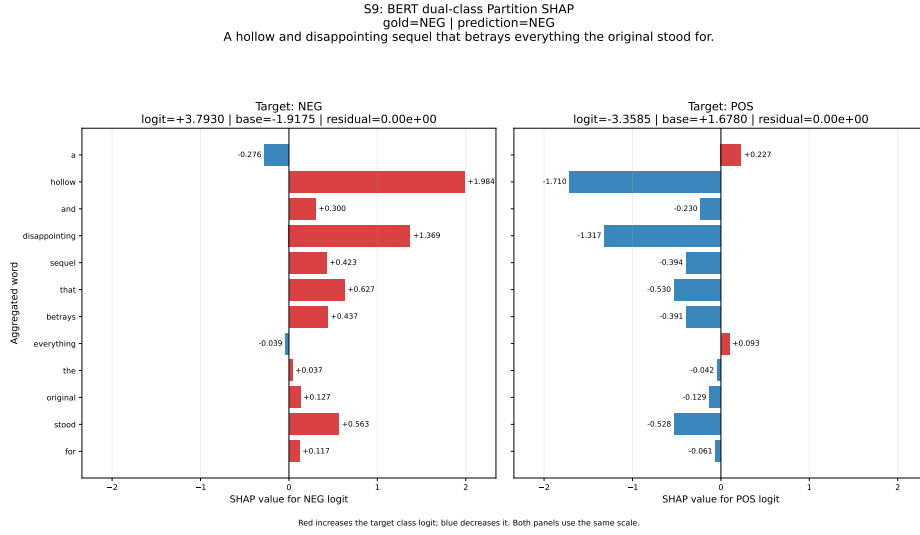


(c) Token-by-class matrix

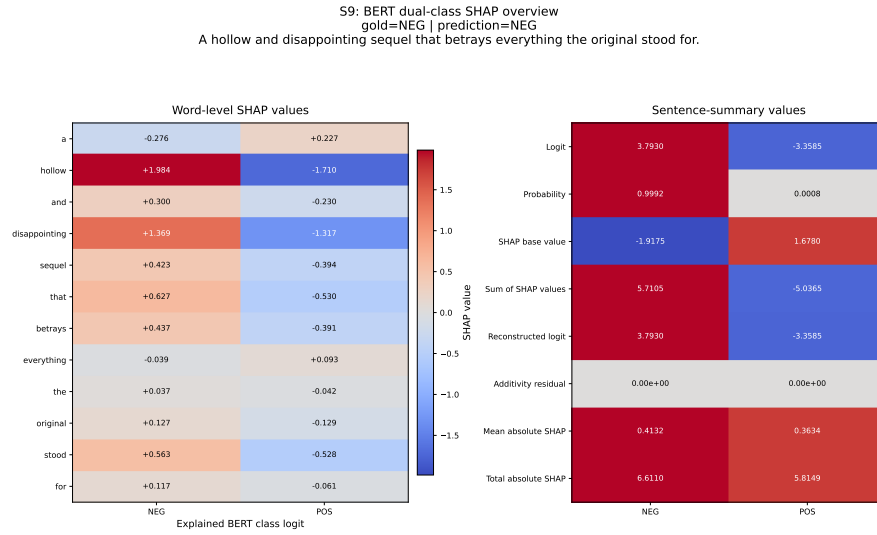
(d) Word-by-class matrix

(e) Sentence summary matrix

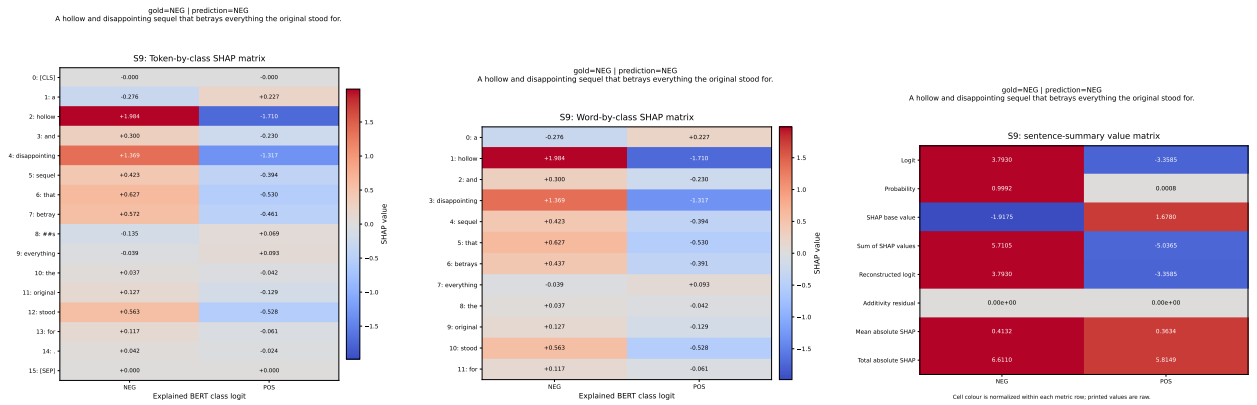
Figure 14: BERT, sentence S8: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

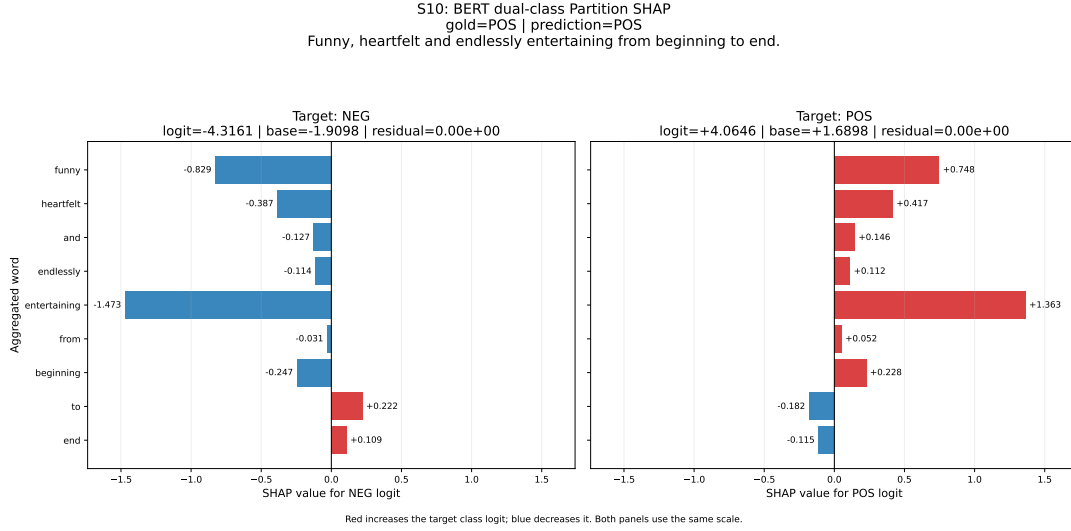


(c) Token-by-class matrix

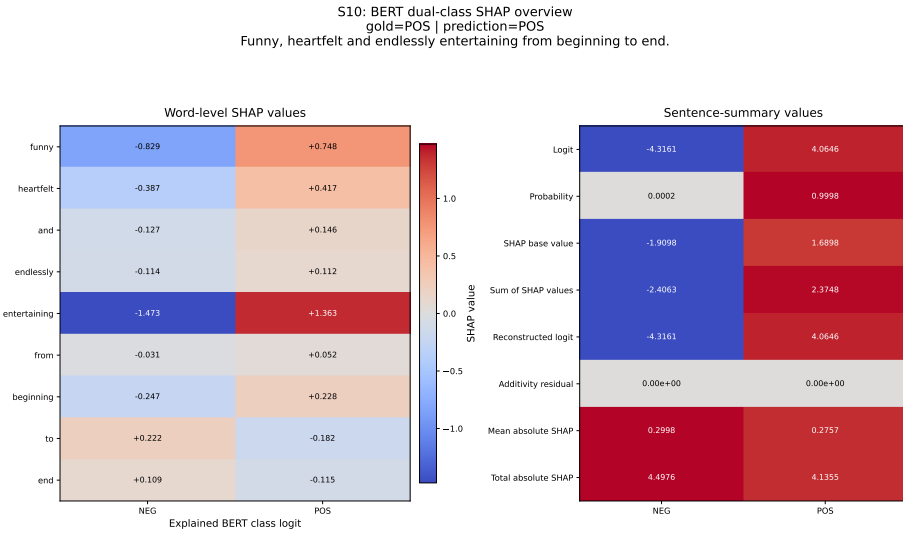
(d) Word-by-class matrix

(e) Sentence summary matrix

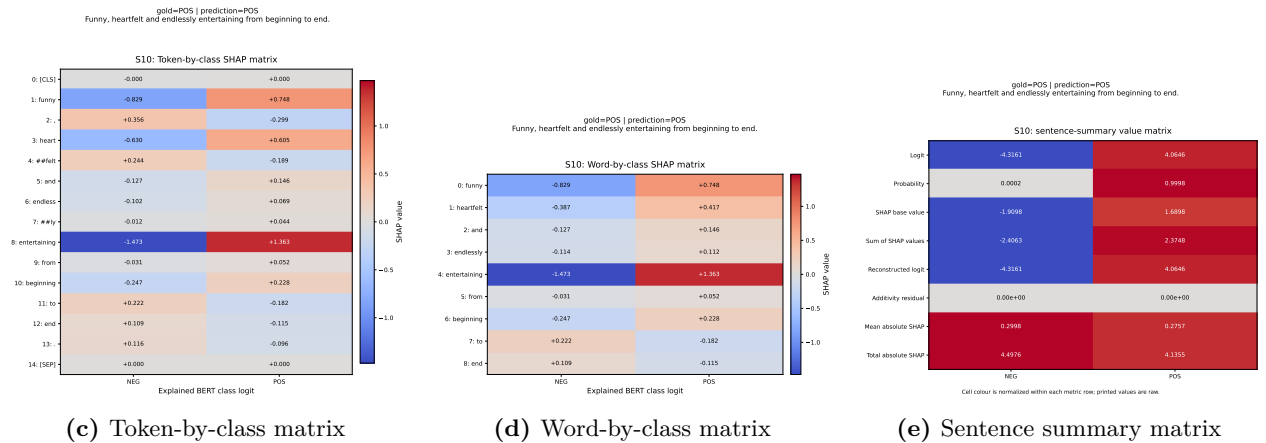
Figure 15: BERT, sentence S9: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary



(c) Token-by-class matrix

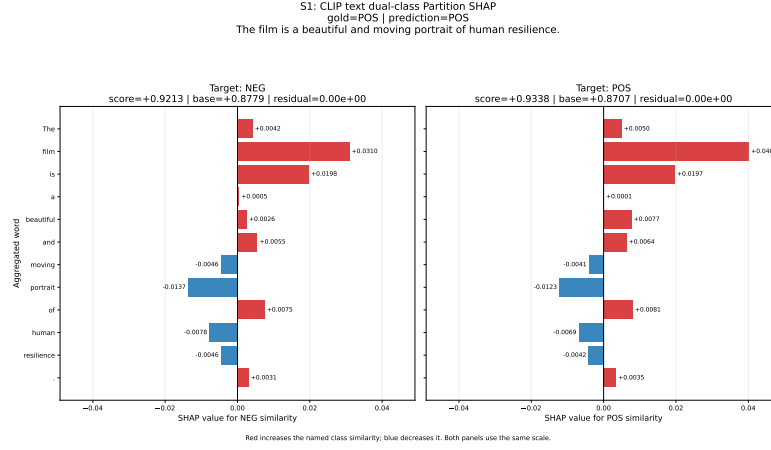
(d) Word-by-class matrix

(e) Sentence summary matrix

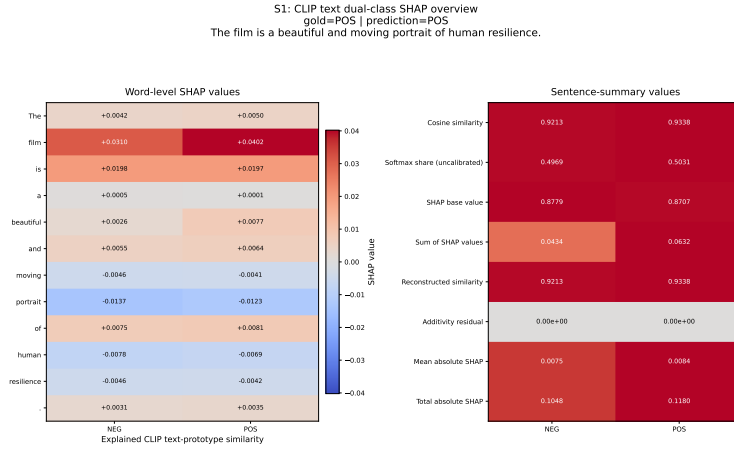
Figure 16: BERT, sentence S10: all five SHAP plots.

G Complete CLIP text sentence-level plots

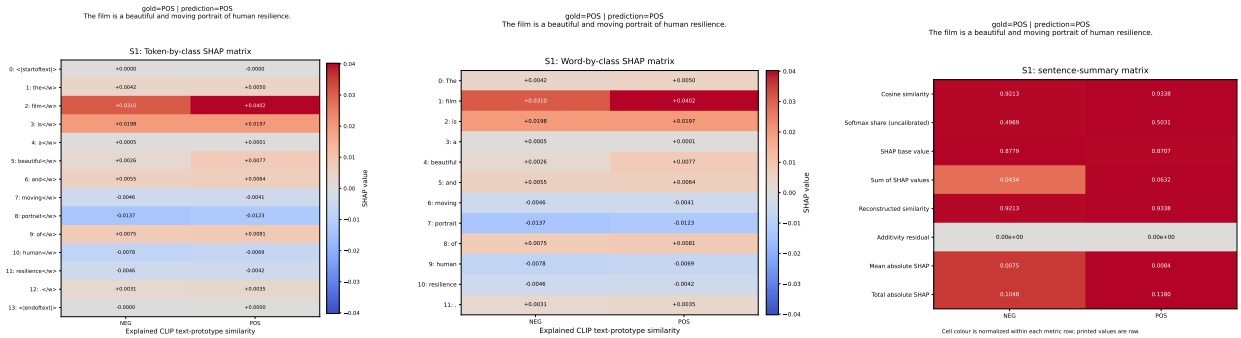
Each page shows the five plots of one sentence. The values explain the cosine similarity to the NEG and POS prompt prototypes; the summary matrix reports the similarities, base values, the sum of the SHAP values, the reconstructed scores and the additivity residual.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

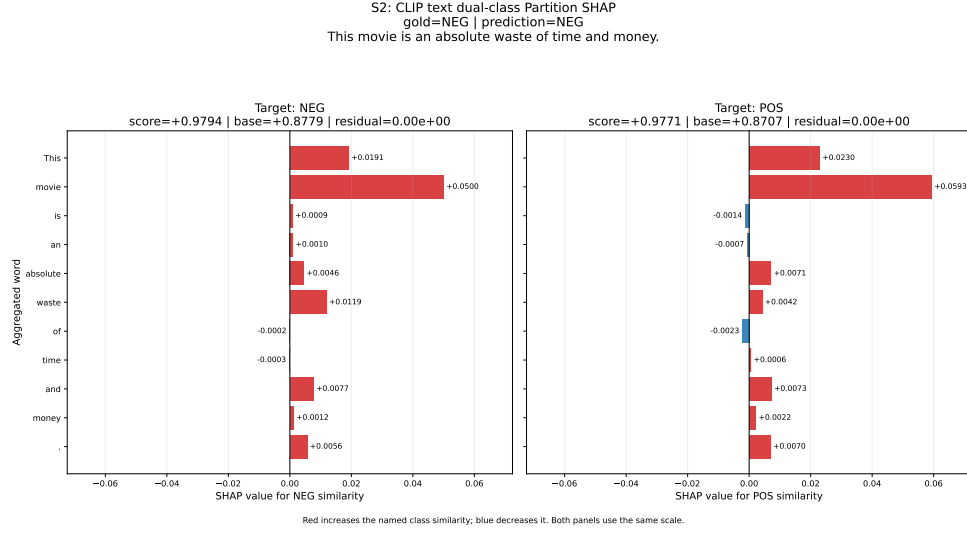


(c) Token-by-class matrix

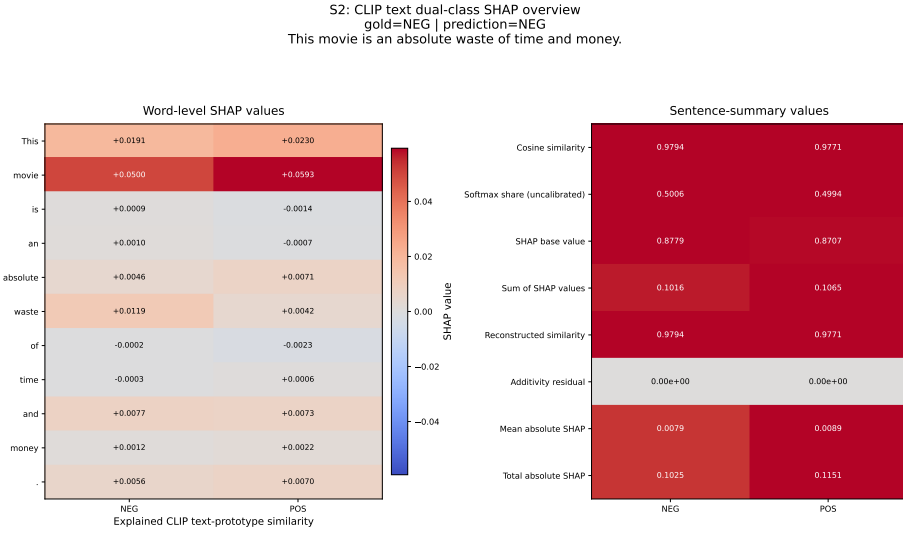
(d) Word-by-class matrix

(e) Sentence summary matrix

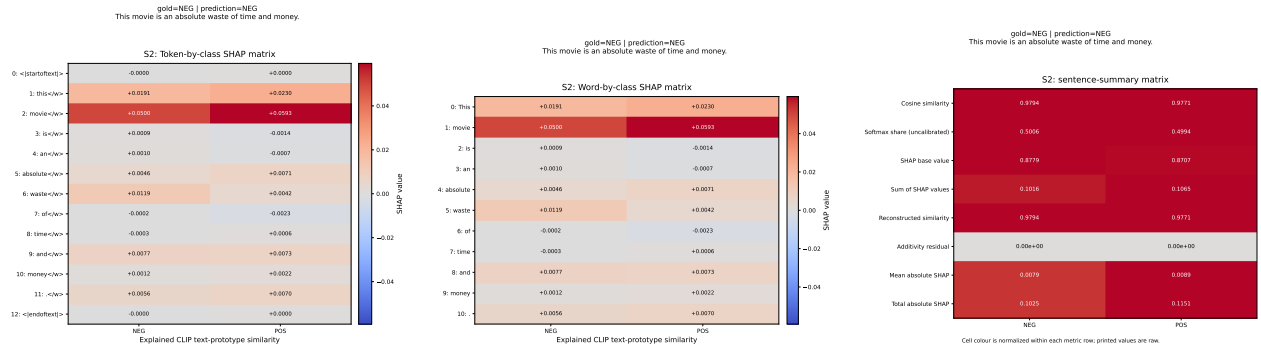
Figure 17: CLIP text, sentence S1: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

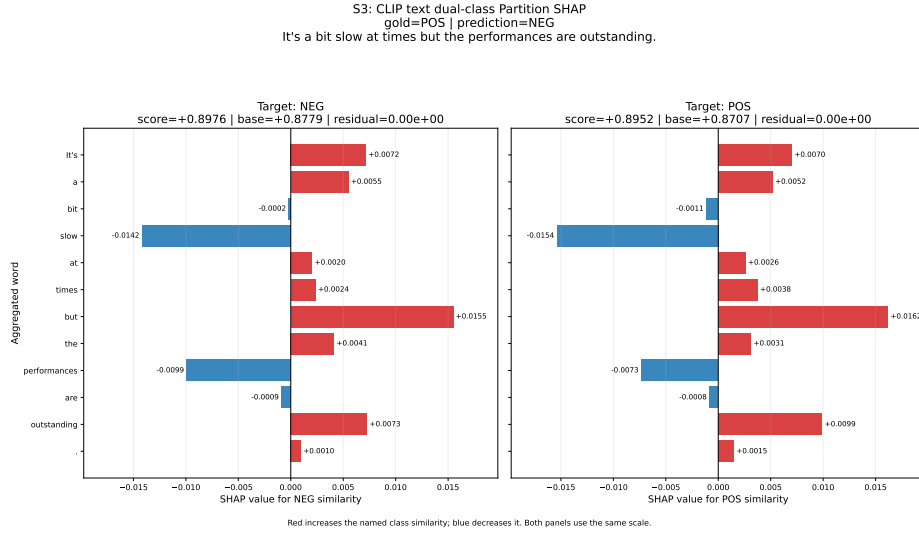


(c) Token-by-class matrix

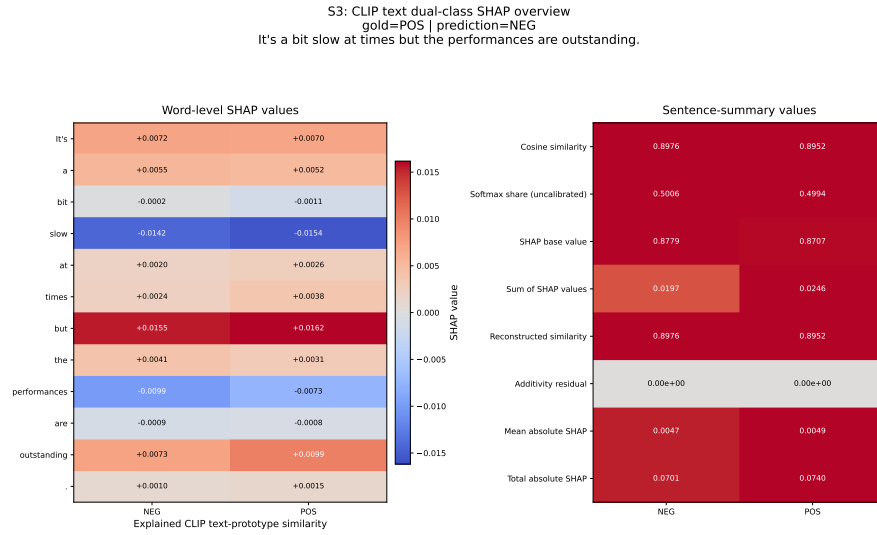
(d) Word-by-class matrix

(e) Sentence summary matrix

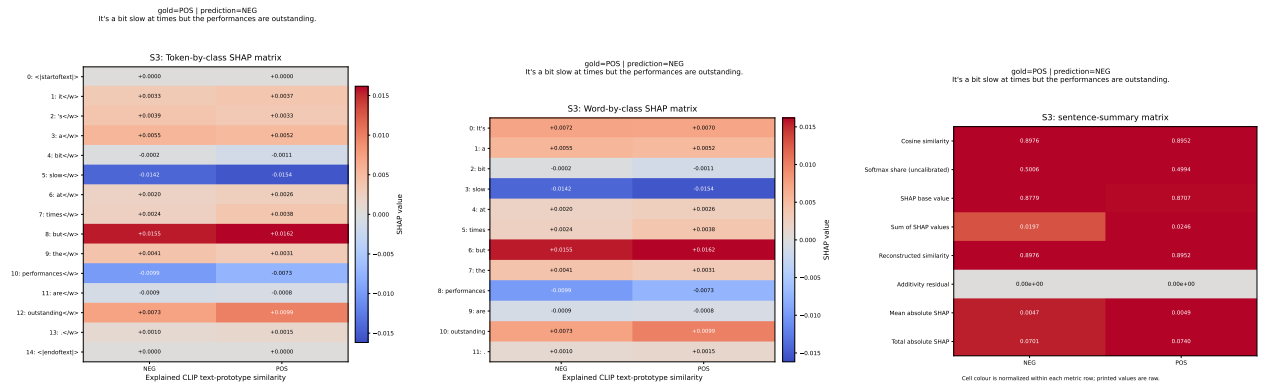
Figure 18: CLIP text, sentence S2: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

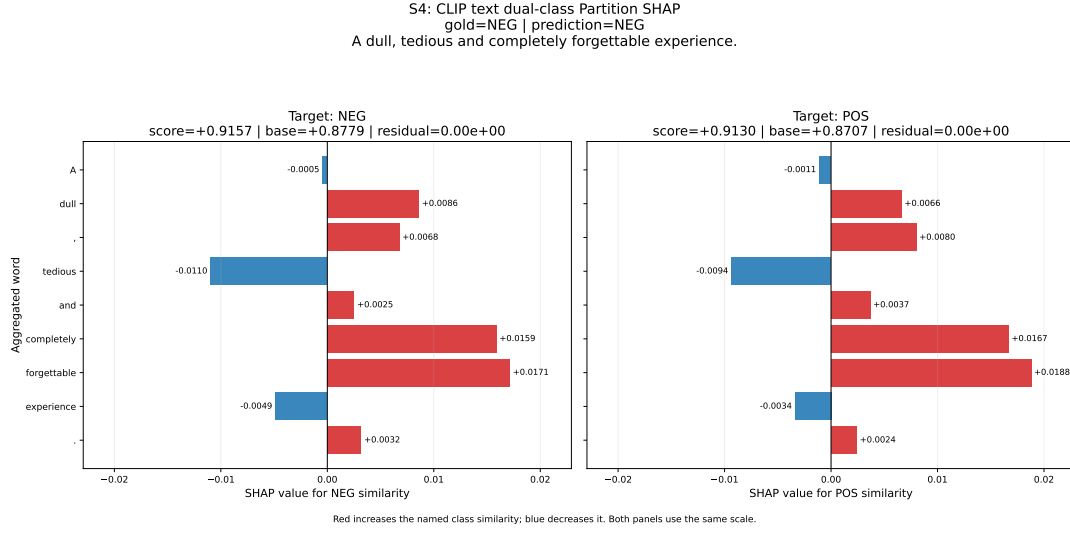


(c) Token-by-class matrix

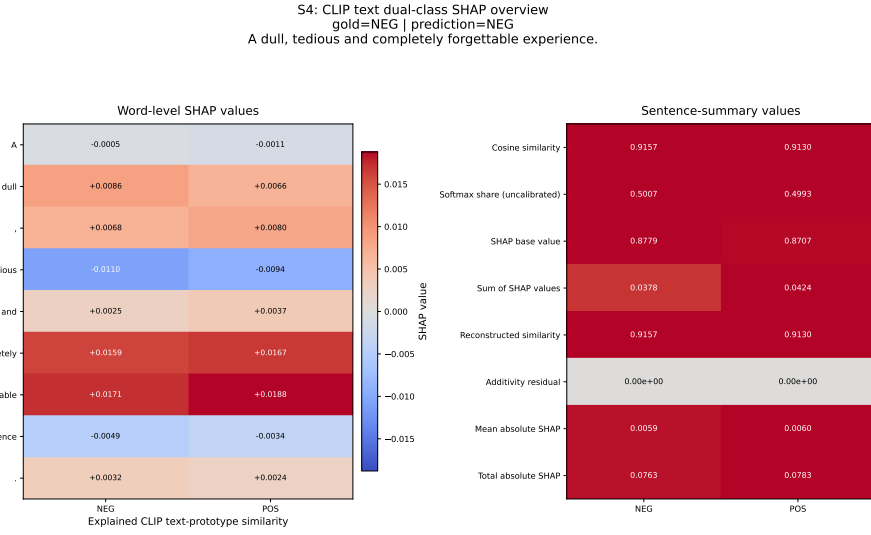
(d) Word-by-class matrix

(e) Sentence summary matrix

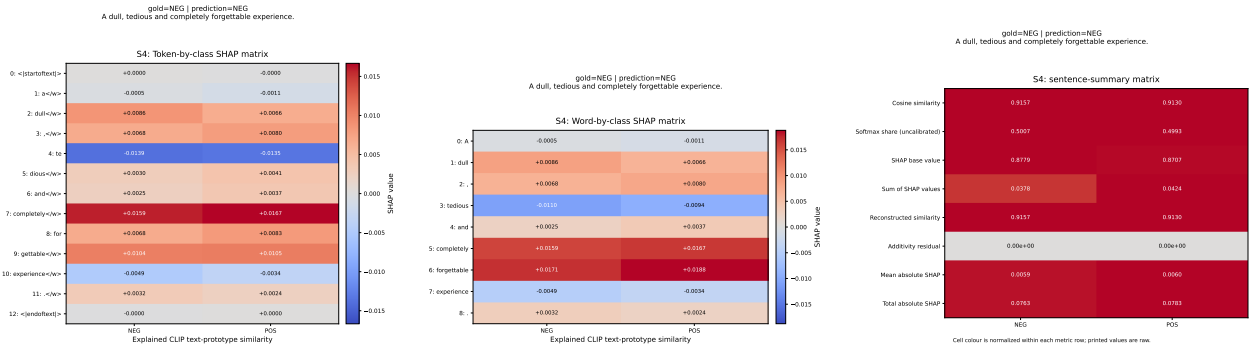
Figure 19: CLIP text, sentence S3: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

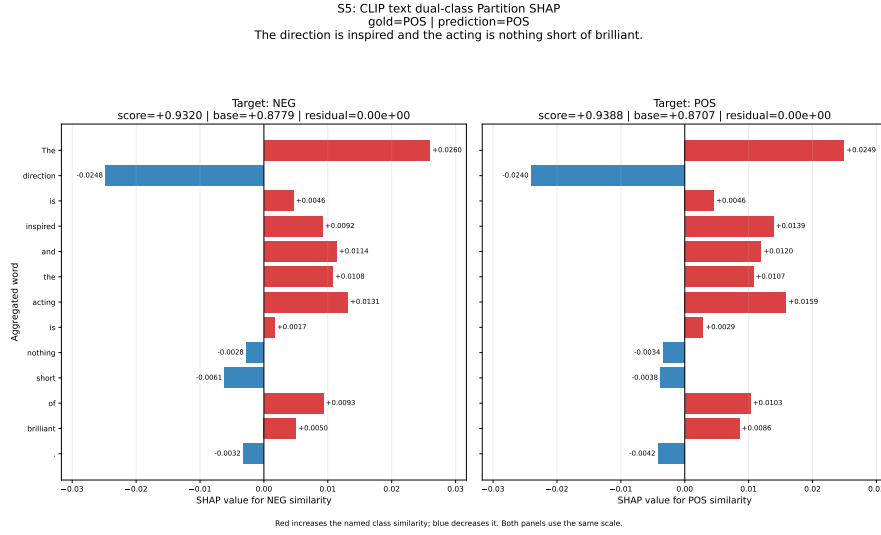


(c) Token-by-class matrix

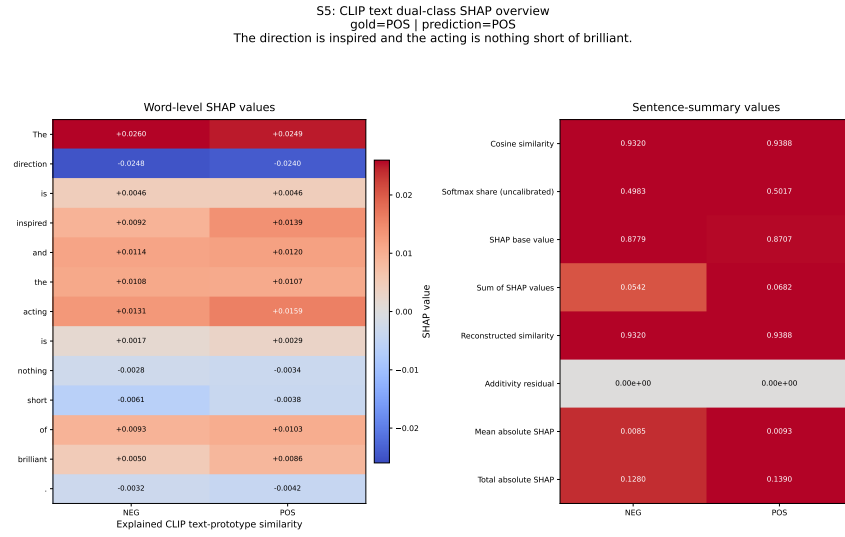
(d) Word-by-class matrix

(e) Sentence summary matrix

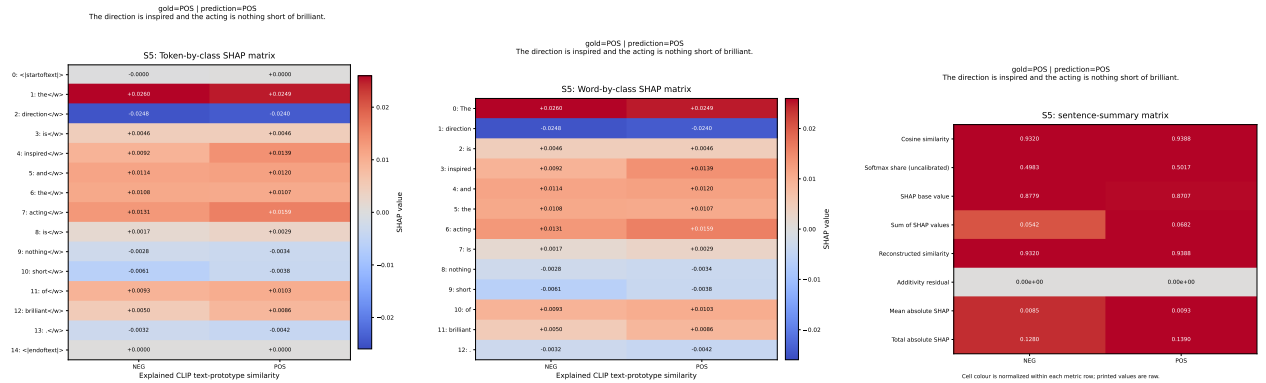
Figure 20: CLIP text, sentence S4: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

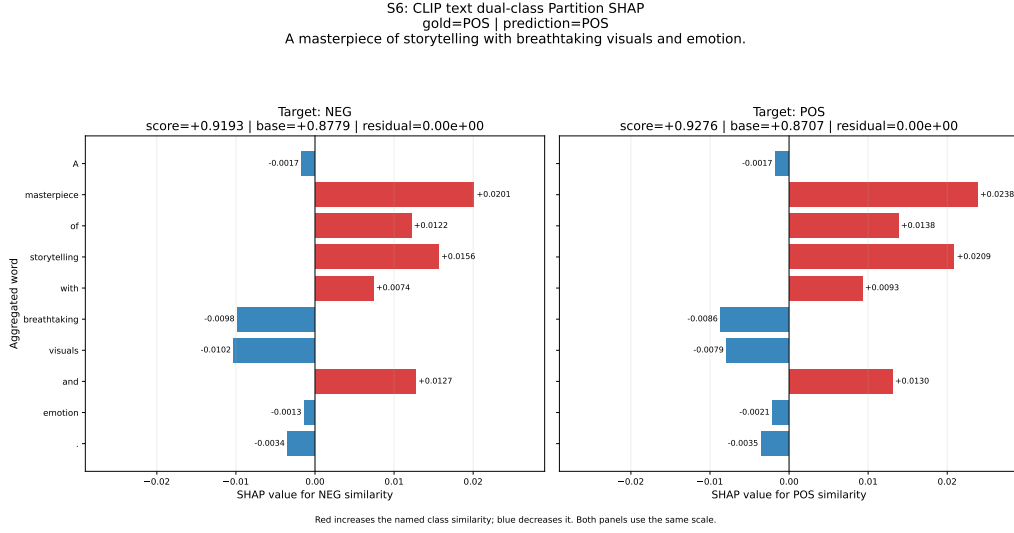


(c) Token-by-class matrix

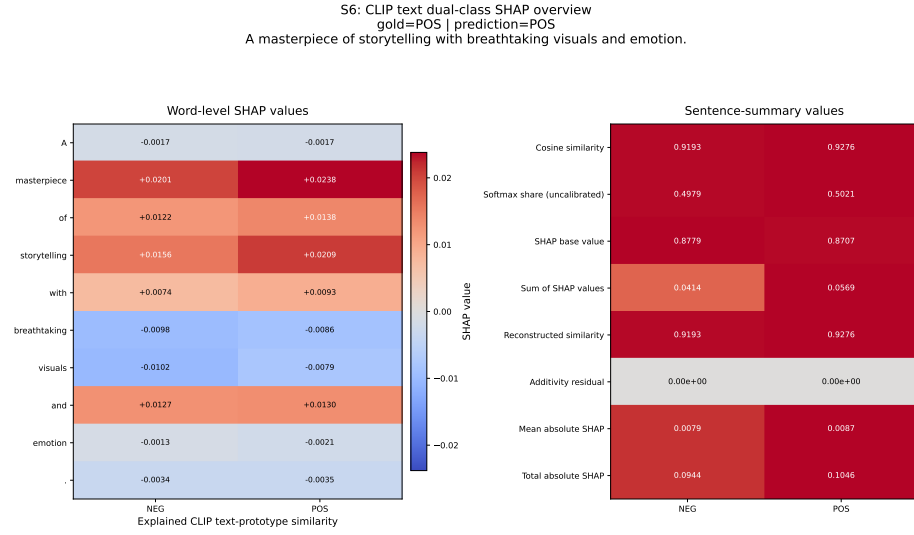
(d) Word-by-class matrix

(e) Sentence summary matrix

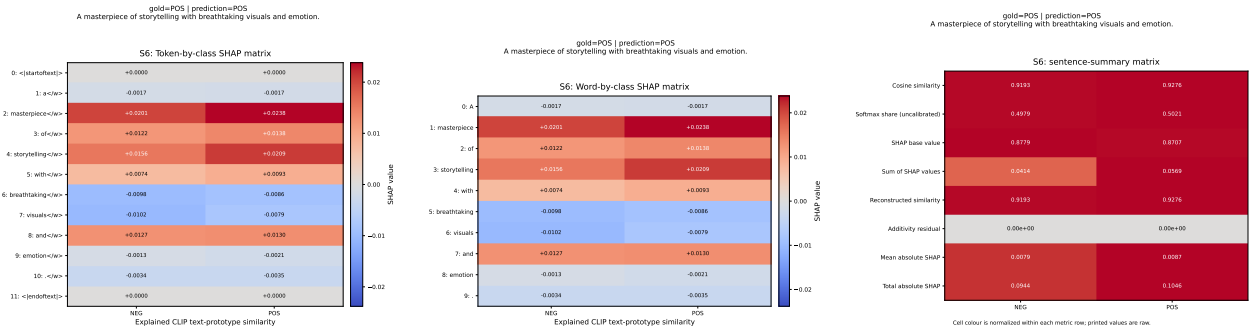
Figure 21: CLIP text, sentence S5: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

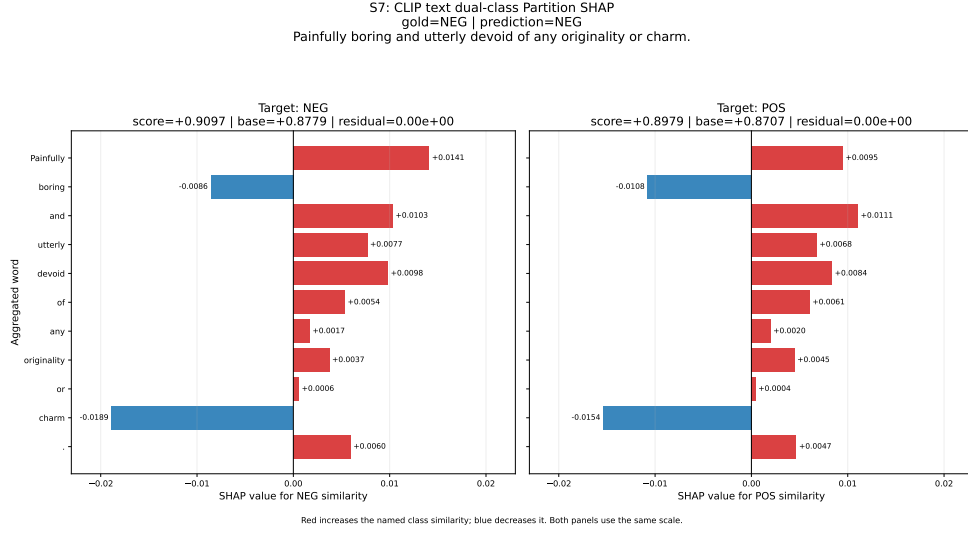


(c) Token-by-class matrix

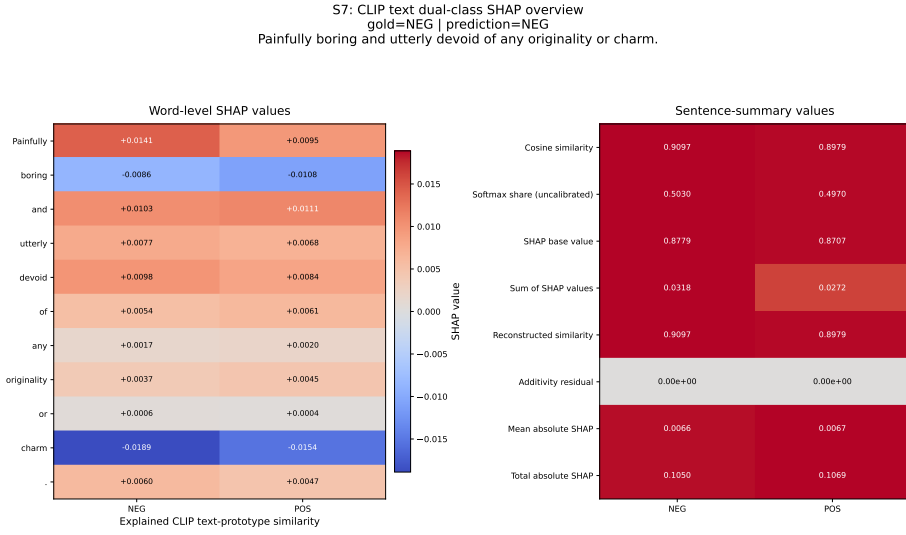
(d) Word-by-class matrix

(e) Sentence summary matrix

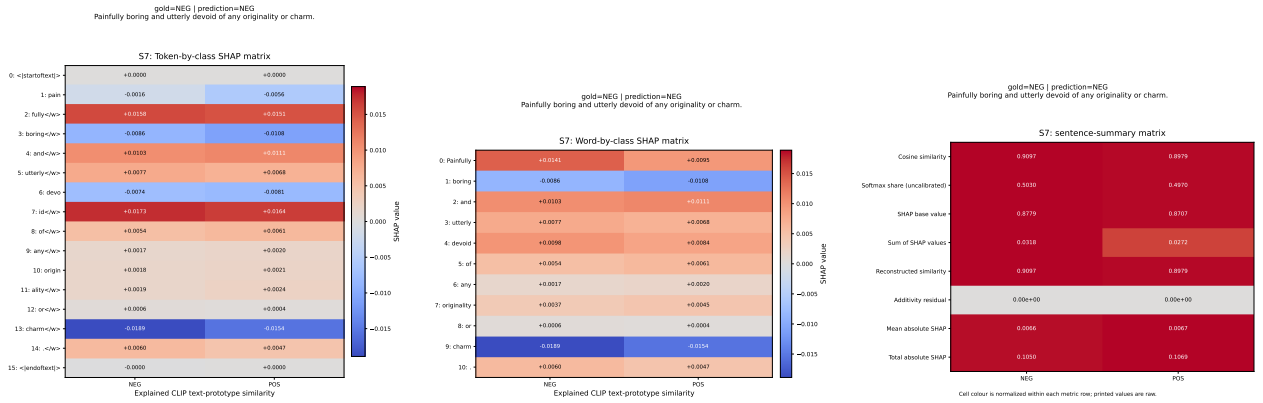
Figure 22: CLIP text, sentence S6: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

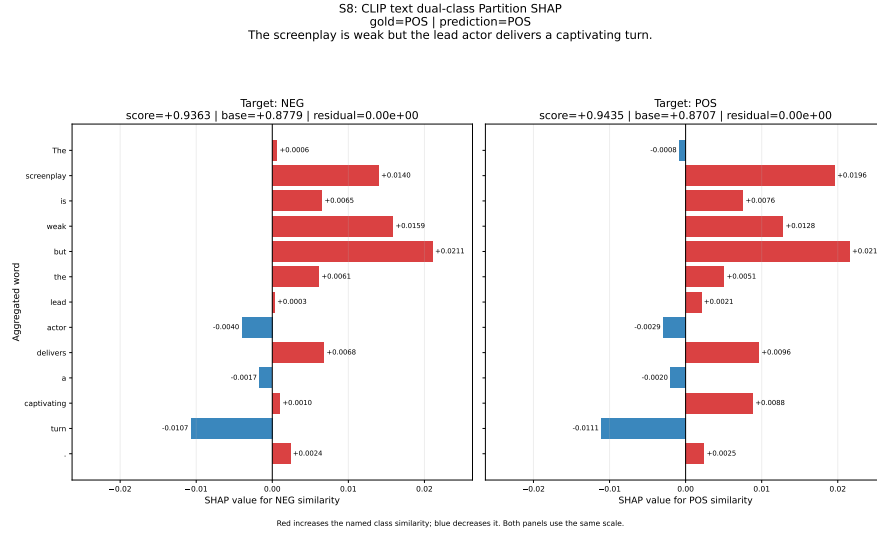


(c) Token-by-class matrix

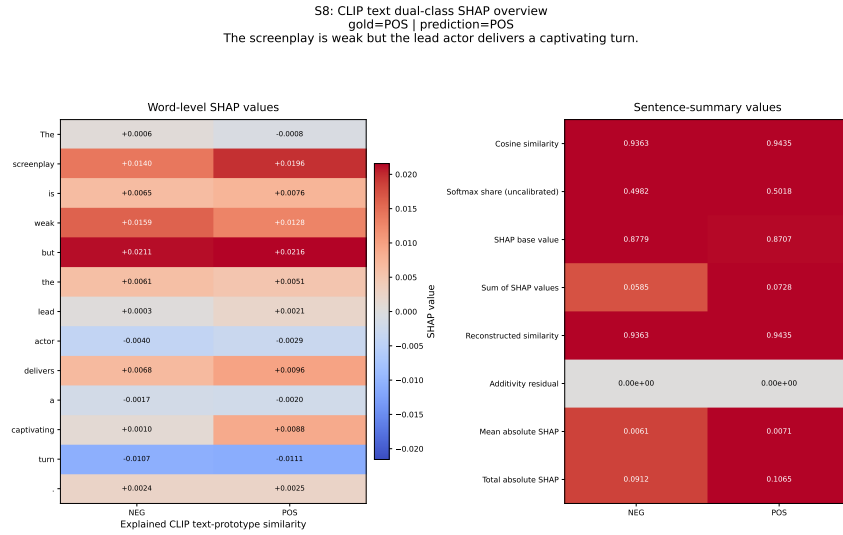
(d) Word-by-class matrix

(e) Sentence summary matrix

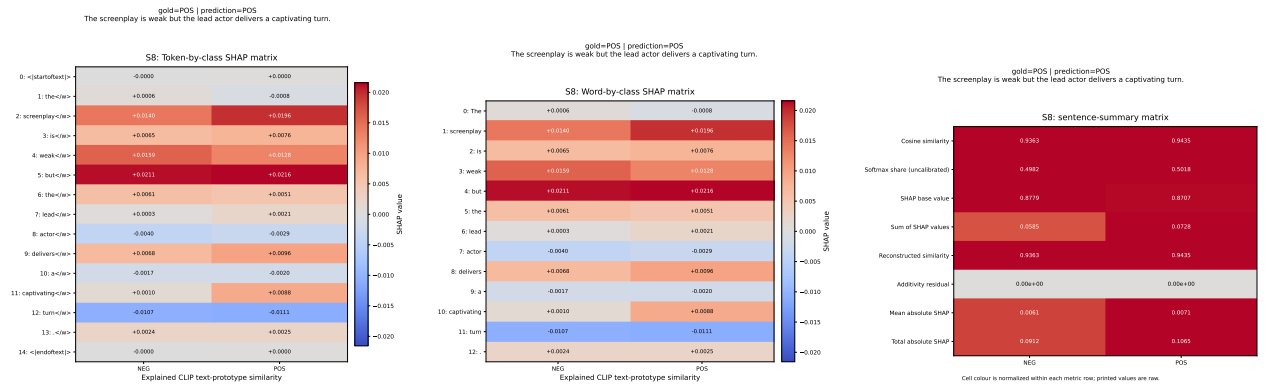
Figure 23: CLIP text, sentence S7: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

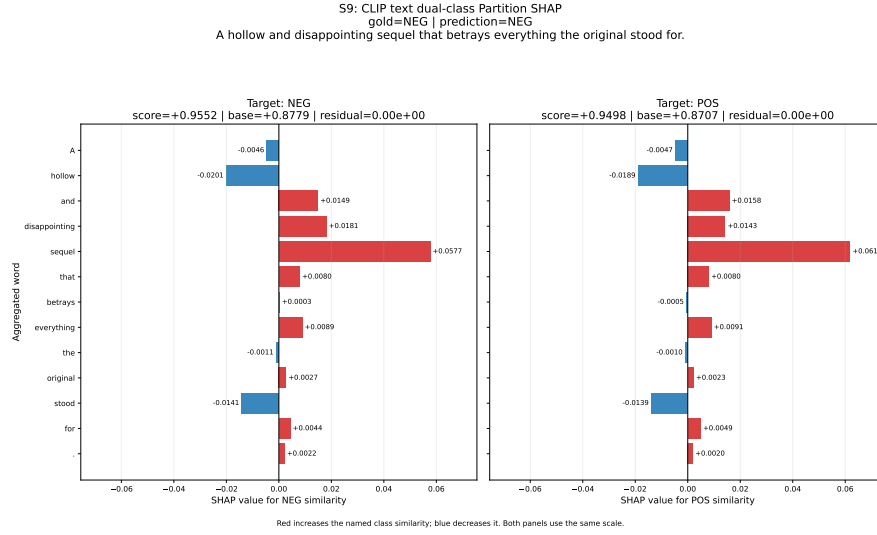


(c) Token-by-class matrix

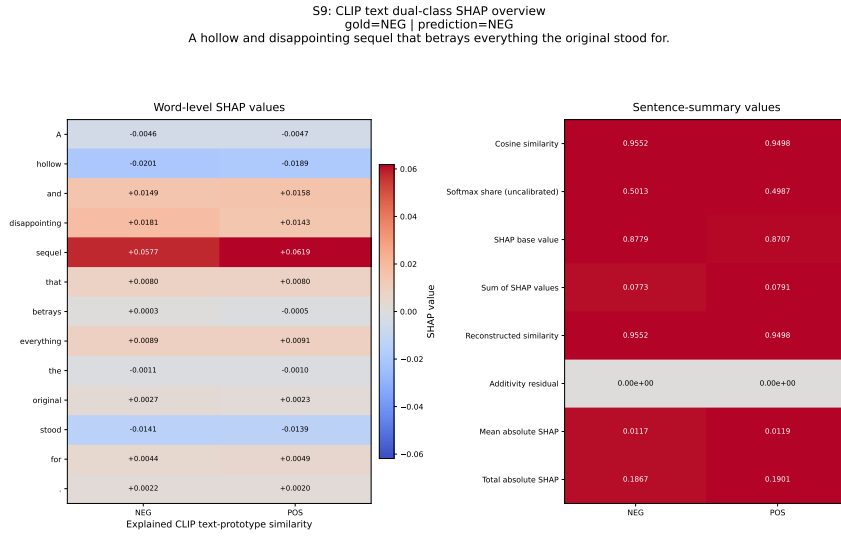
(d) Word-by-class matrix

(e) Sentence summary matrix

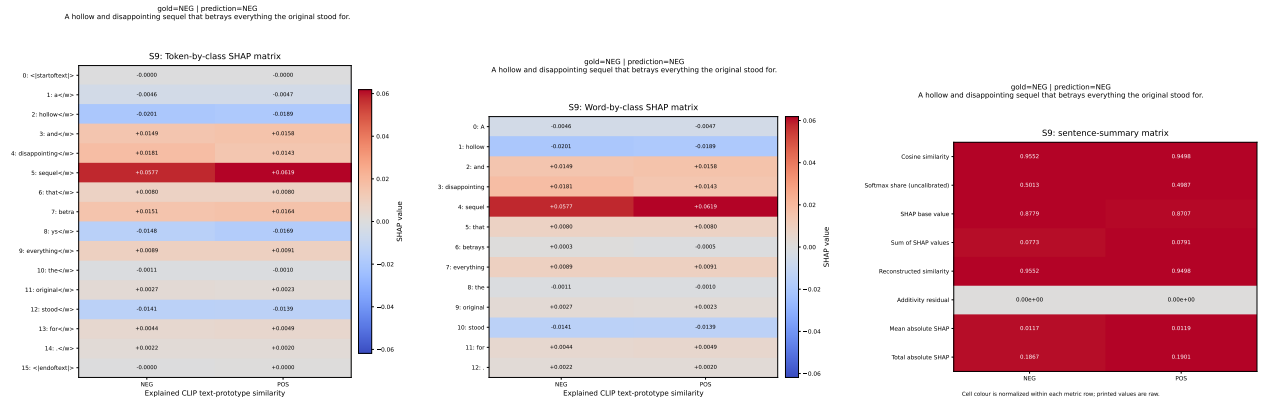
Figure 24: CLIP text, sentence S8: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary

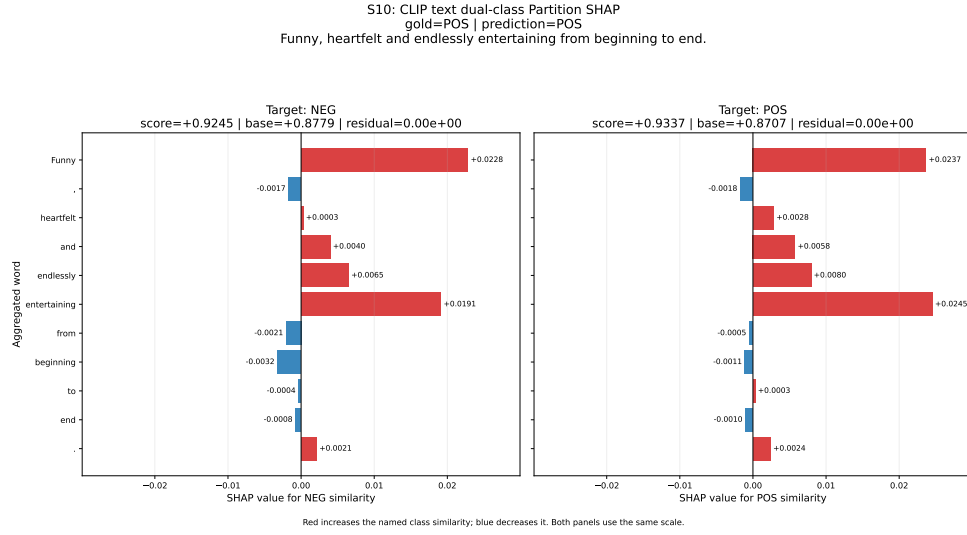


(c) Token-by-class matrix

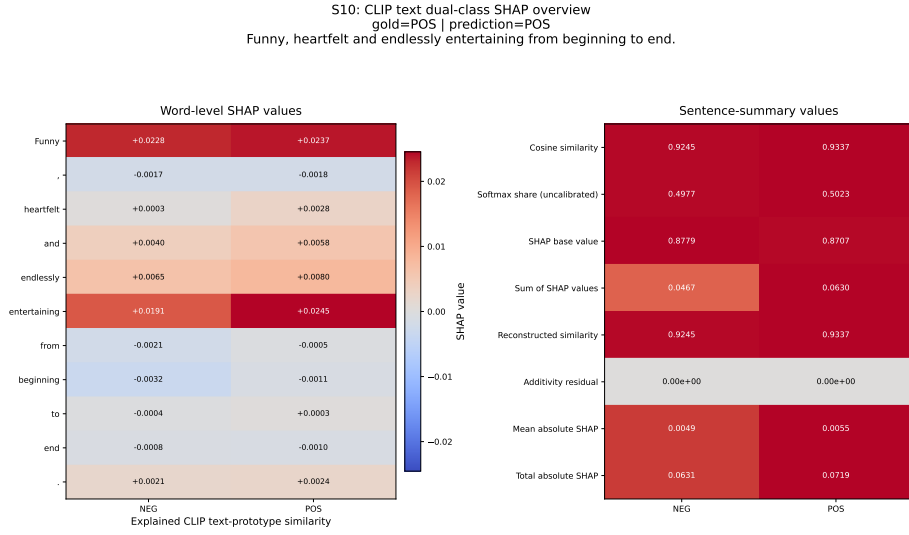
(d) Word-by-class matrix

(e) Sentence summary matrix

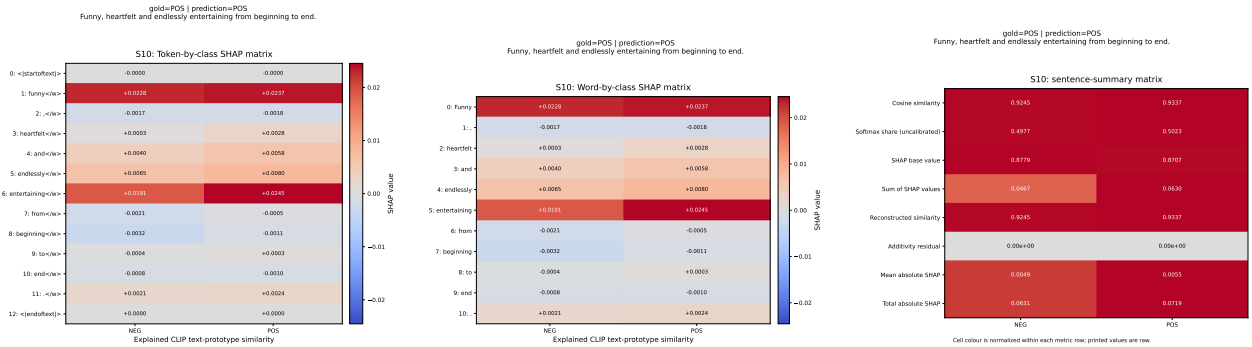
Figure 25: CLIP text, sentence S9: all five SHAP plots.



(a) Word-level SHAP bars for the NEG and POS outputs



(b) Presentation overview: word values and sentence summary



(c) Token-by-class matrix

(d) Word-by-class matrix

(e) Sentence summary matrix

Figure 26: CLIP text, sentence S10: all five SHAP plots.

H Complete CLIP vision patch-level plots

Each figure shows the 7×7 patch SHAP values over the 224×224 input that CLIP sees, with the value printed in each patch. The title reports the score of the original image (1), the base value (the fully blurred image) and the additivity residual.

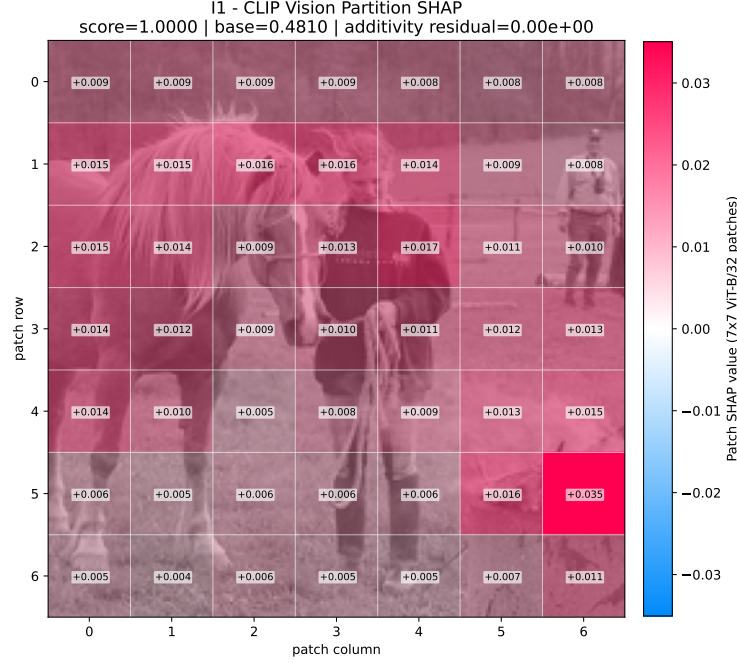


Figure 27: CLIP vision, image I1: 7×7 patch SHAP values over the 224×224 model input.

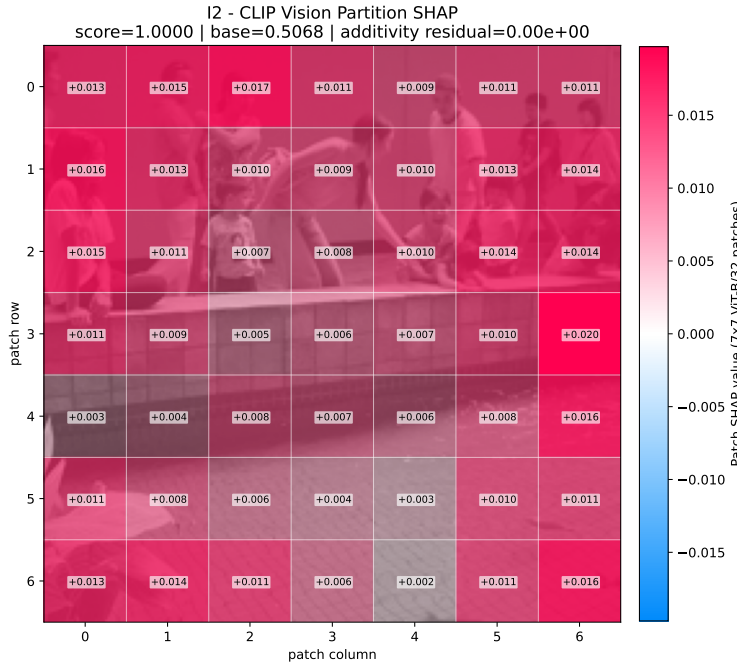


Figure 28: CLIP vision, image I2: 7×7 patch SHAP values over the 224×224 model input.

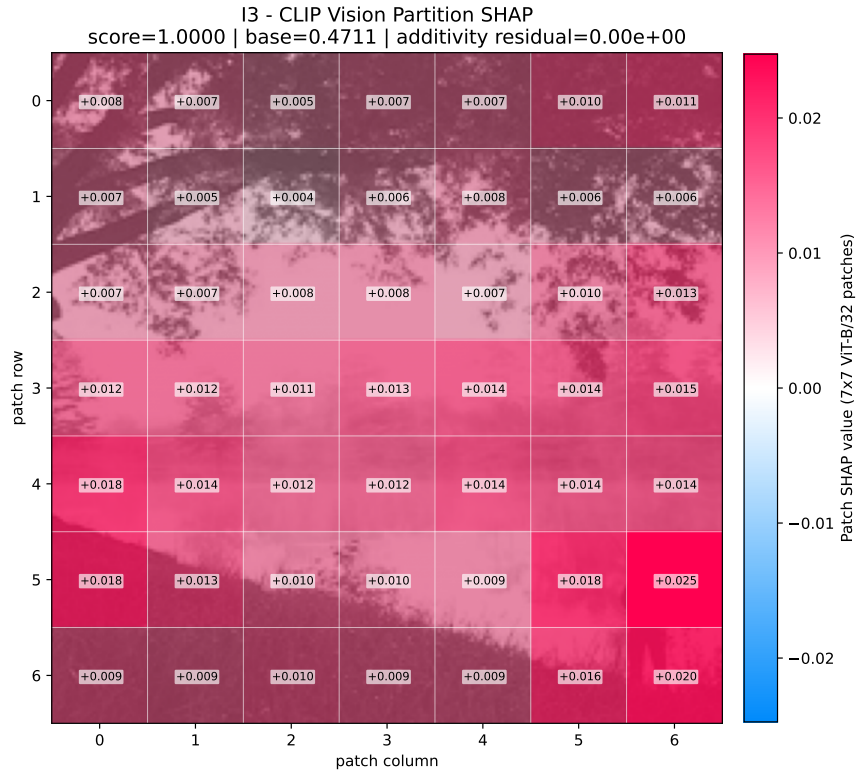


Figure 29: CLIP vision, image I3: 7×7 patch SHAP values over the 224×224 model input.

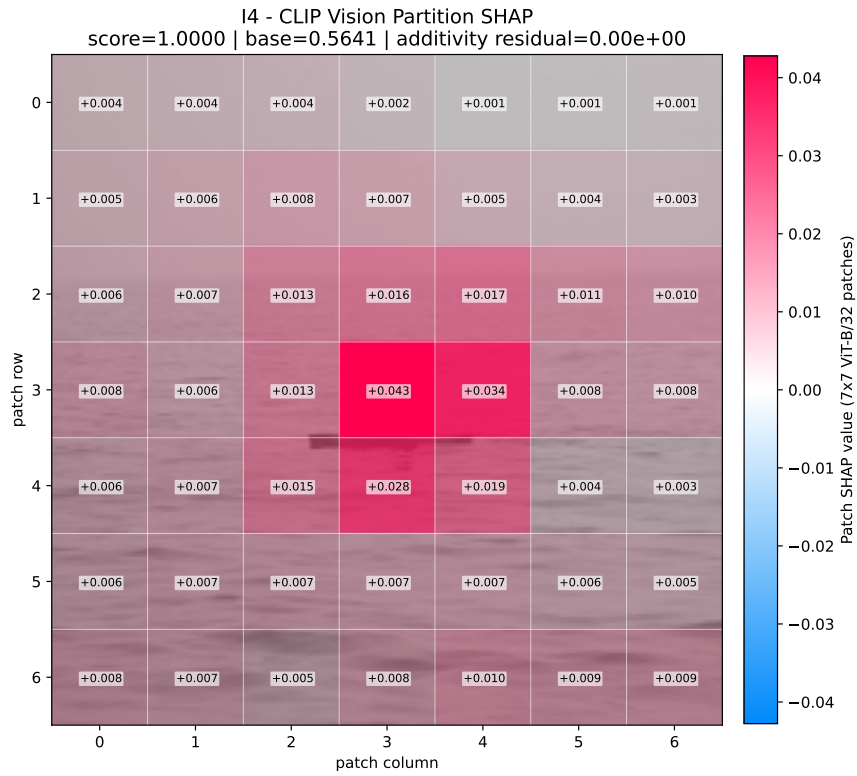


Figure 30: CLIP vision, image I4: 7×7 patch SHAP values over the 224×224 model input.

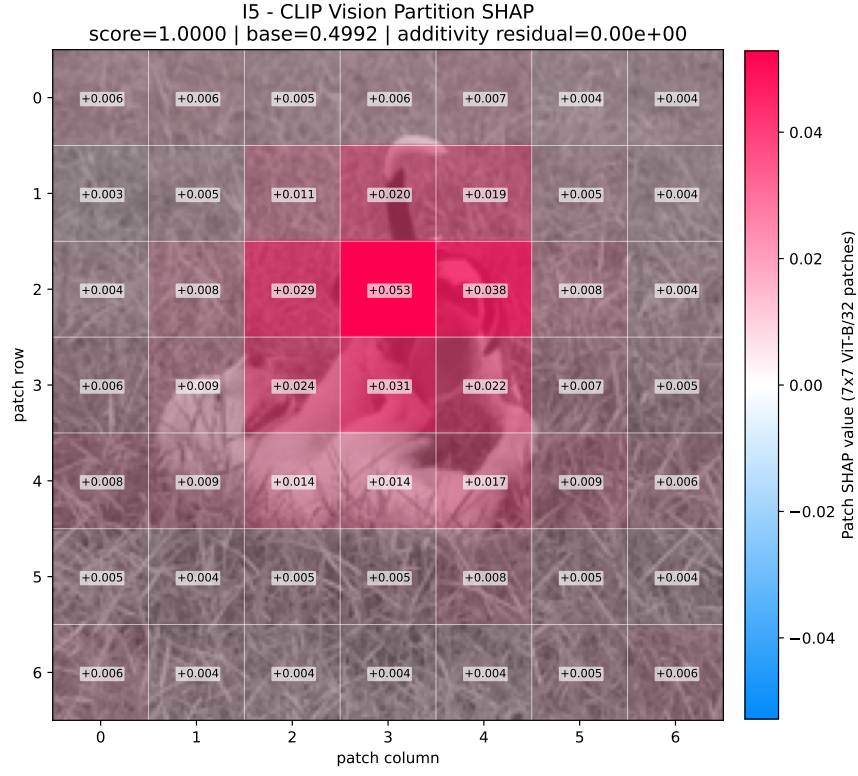


Figure 31: CLIP vision, image I5: 7×7 patch SHAP values over the 224×224 model input.

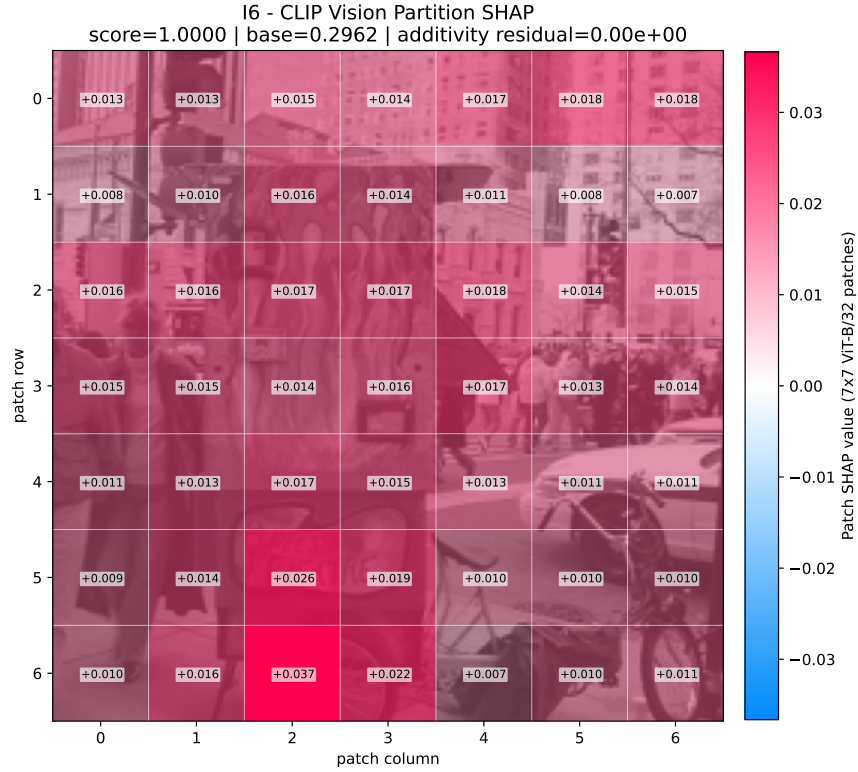


Figure 32: CLIP vision, image I6: 7×7 patch SHAP values over the 224×224 model input.

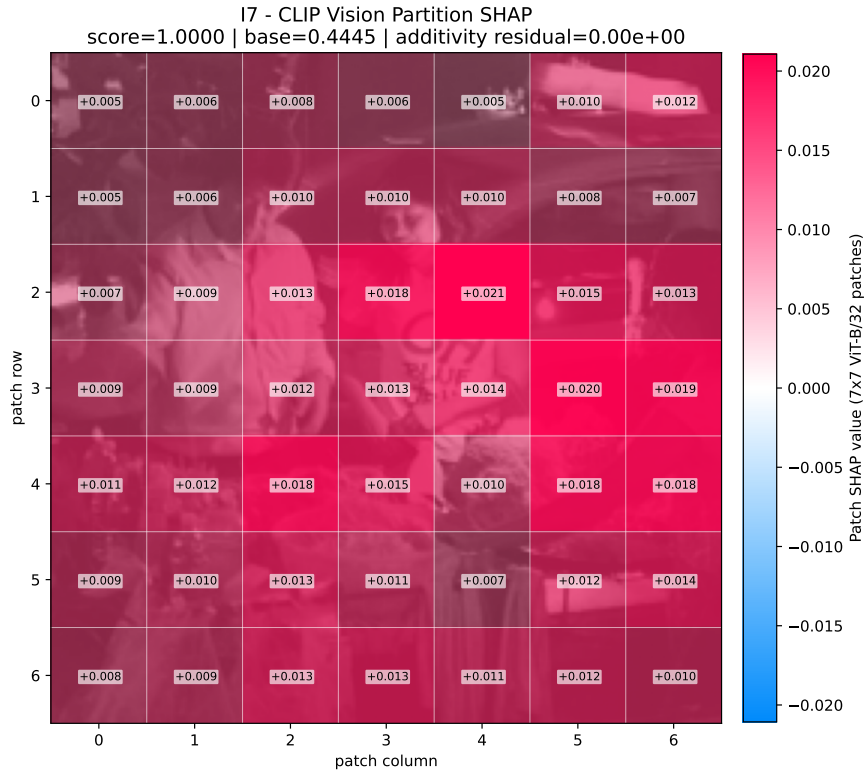


Figure 33: CLIP vision, image I7: 7×7 patch SHAP values over the 224×224 model input.

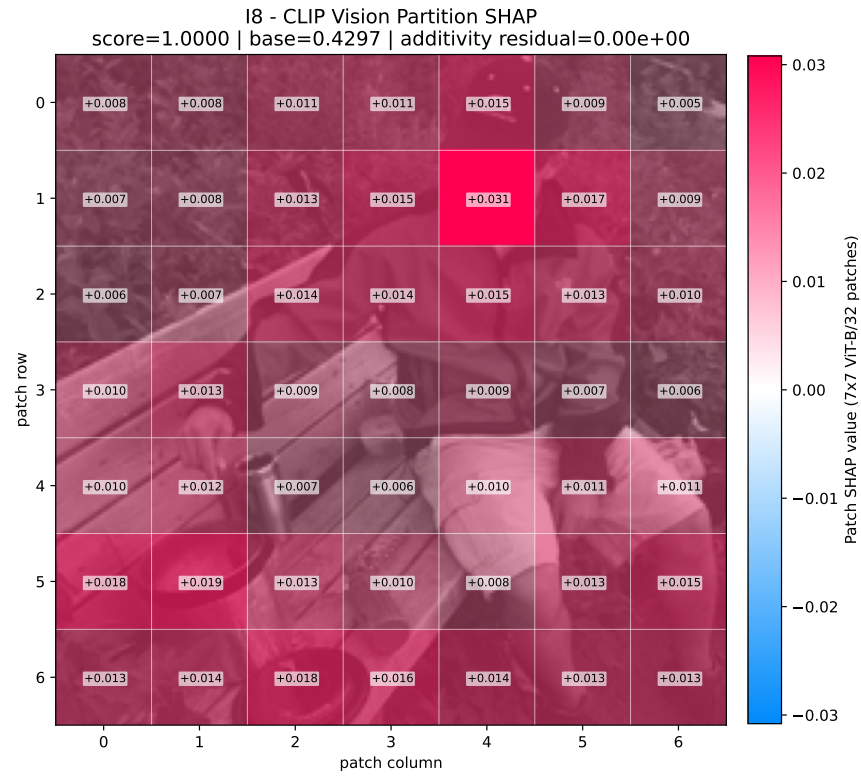


Figure 34: CLIP vision, image I8: 7×7 patch SHAP values over the 224×224 model input.

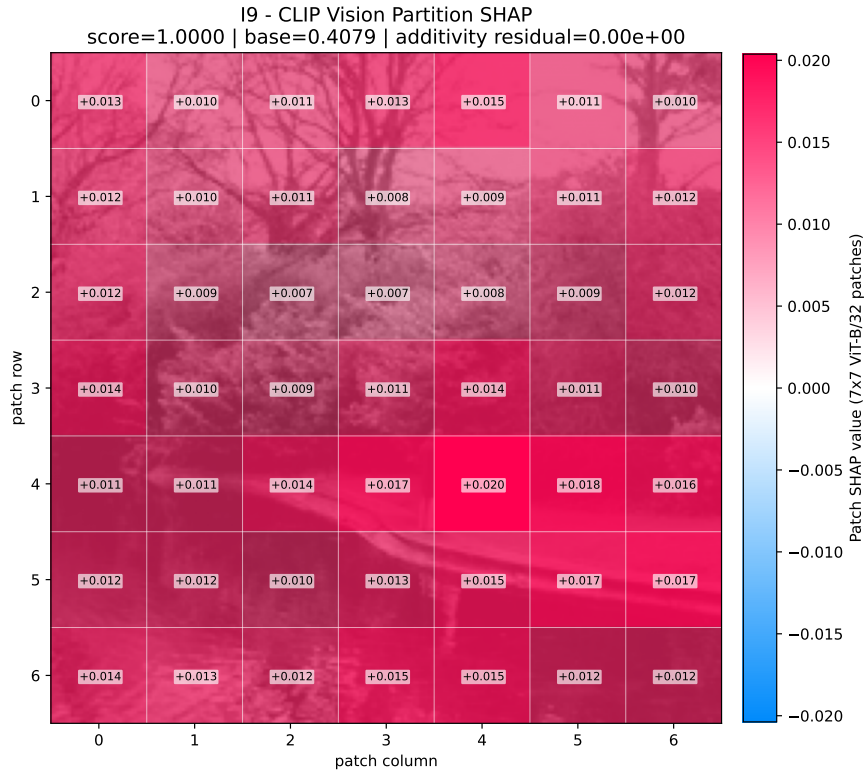


Figure 35: CLIP vision, image I9: 7×7 patch SHAP values over the 224×224 model input.

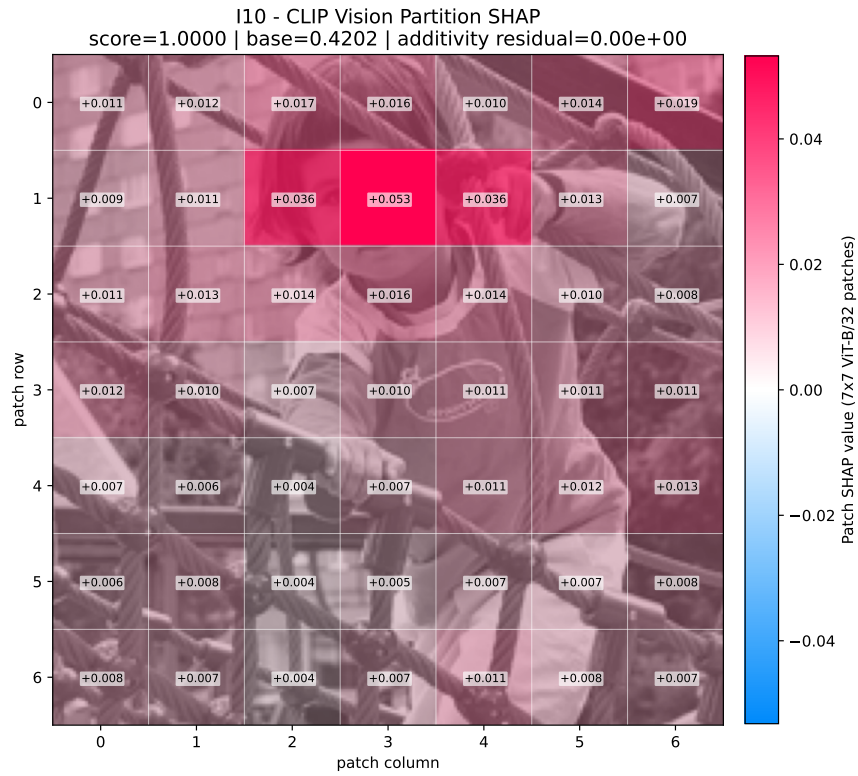


Figure 36: CLIP vision, image I10: 7×7 patch SHAP values over the 224×224 model input.